

Data dossier

Prediction of Dynamic Stall on a Pitching NACA 0012 Section

A case study at a tunnel condition and at a rotor-blade-station condition

UNISTALL, a Leishman-Beddoes dynamic-stall load model with tabulated separation (Python package unistall)
Leishman-Beddoes indicial method with a tabulated separation point

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The data behind the report, as tables. Each table names the file it is read from; no number is typed by hand.

The loads are the prediction of the load model. The flow fields listed in the inventory are a potential-flow reconstruction, incompressible with a linearised correction of the pressure, quasi-steady, not a flow solution.

Case B is an aerofoil at a rotor-blade-station condition in a steady stream with prescribed pitch, not a rotor calculation.

1 Flow conditions

Source: 03_model_setup/flow_conditions.csv. 12 rows.

parameter	A validation	B application	units	source
description	tunnel condition of NASA TM-84245, frame 9302	an aerofoil at a rotor-blade-station condition (steady stream, prescribed pitch); not a rotor calculation	-	project_meta.CASES
airfoil	NACA 0012	NACA 0012	-	project_meta.SECTION
chord_c	0.61	0.527	m	Case A: NASA TM-84245; Case B: Bousman, NASA/TP-2003-212265 (UH-60A main rotor)
freestream_mach_M	0.3	0.28	-	Case A: nominal condition of the frame; Case B: station_condition.csv
speed_of_sound_a	340.29228686527705	340.29228686527705	m/s	sea-level standard air, sqrt(gamma R T)
freestream_velocity_U	102.0876860595831	95.28184032227759	m/s	M a
air_density_rho	1.22501	1.22501	kg/m ³	sea-level standard air
static_pressure_p_inf	101325.0	101325.0	Pa	sea-level standard air
static_temperature_T_inf	288.15	288.15	K	sea-level standard air
dynamic_viscosity_mu	1.7893 × 10 ⁻⁵	1.7893 × 10 ⁻⁵	Pa.s	sea-level standard air, Sutherland
reynolds_number_Re_c	4263000.0	3438000.0	-	sea-level standard air; not the tunnel value of the frame and not a flight value; the load model does not use it
reduced_frequency_k	0.1	0.07471689346280729	-	Case A: nominal condition of the frame; Case B: station_condition.csv

2 Kinematics

Source: 03_model_setup/kinematics.csv. 2 rows.

case	alpha mean deg	alpha amp deg	reduced freq k	omega rad s	freq Hz	period s	motion
A_validation	10	10	0.1	33.471372	5.327134	0.187718	alpha(t) = mean + amp sin(omega t), about the quarter chord
B_application	12	8	0.07471689346	27.017697	4.3	0.232558	alpha(t) = mean + amp sin(omega t), about the quarter chord

3 How the condition of Case B follows from the rotor (Case B is not a rotor calculation)

Source: 03_model_setup/station_condition.csv. 23 rows.

quantity	value	units	source
rotor_radius	26.83	ft	Bousman, NASA/TP-2003-212265 (UH-60A main rotor)
rotor_radius_m	8.1778	m	radius in feet x 0.3048
blade_chord	20.76	in	Bousman, NASA/TP-2003-212265 (UH-60A main rotor)
blade_chord_m	0.527	m	chord in inches x 0.0254, to three decimals
number_of_blades	4	-	Bousman, NASA/TP-2003-212265 (UH-60A main rotor)
rotor_speed	258	rpm	Bousman, NASA/TP-2003-212265 (UH-60A main rotor)
rotor_speed_rev_per_s	4.3	1/s	rpm / 60
rotor_speed_Omega	27.0177	rad/s	2 pi rpm / 60
tip_speed	220.945	m/s	Omega R
station_r_over_R	0.75	-	chosen for this study
station_rotational_speed	165.709	m/s	Omega r
station_azimuth	270.0	deg	chosen for this study: the retreating side
station_mach_M	0.28	-	chosen for this study: the Mach number of Case B
speed_of_sound	340.292	m/s	sea-level standard air, sqrt(gamma R T)
station_speed_U	95.282	m/s	M a
forward_speed_implied	70.43	m/s	the forward speed V at which $\Omega r + V \sin(\text{azimuth}) = U$
forward_speed_implied_kt	136.9	kt	V in knots
advance_ratio_implied	0.3188	-	$V / (\Omega R)$
mach_at_station_lowest	0.28	-	$(\Omega r - V) / a$: the value Case B holds steady
mach_at_station_highest	0.6939	-	$(\Omega r + V) / a$, on the advancing side: not represented in Case B
pitch_frequency	4.3	Hz	once per revolution: rpm / 60
reduced_frequency_k	0.074717	-	$\Omega c / (2 U)$
incidence_history	12 deg +/- 8 deg, sinusoidal	-	assumed for this study; it comes from no trim calculation

4 Air properties

Source: 03_model_setup/air_properties.csv. 10 rows.

property	value	units	note
gamma	1.4	-	ratio of specific heats
R_gas	287.05	J/kg/K	specific gas constant
prandtl	0.72	-	Prandtl number of air
T_inf	288.15	K	sea-level standard temperature
p_inf	101325	Pa	sea-level standard pressure
a_sound	340.2922869	m/s	derived: $\sqrt{\text{gamma R T}}$
rho	1.22501	kg/m ³	derived: $p / (R T)$
mu	1.7893×10^{-5}	Pa.s	derived: Sutherland's law
cp	1004.675	J/kg/K	derived: $\text{gamma R} / (\text{gamma} - 1)$
recovery_factor	0.8963	-	derived: $\text{Pr}^{(1/3)}$, the usual value for a turbulent boundary layer on a flat plate, with $\text{Pr} = 0.72$ for air

5 Solver configuration

Source: 03_model_setup/solver_config.json. 118 rows.

Key	Value
author	Akosa Samuel Onyejekwe (independent)
title	Prediction of Dynamic Stall on a Pitching NACA 0012 Section
solver_name	UNISTALL
solver	UNISTALL, a Leishman-Beddoes dynamic-stall load model with tabulated separation (Python package unistall)
method	Leishman-Beddoes indicial method with a tabulated separation point
package	unistall
load_model	unistall.dsmodel.solve
section	0012
constants.Tp	1.651526
constants.Tf0	6.278239

Key	Value
constants.Tv0	6.317996
constants.Tvl	8.391549
constants.eta	0.906958
constants.xcp_v	0.2
constants.St_sh	0.19
constants.cm_lag	0.1
constants.CD0	0.0072
constants.cn1_factor	1
constants.cmq_kirchhoff	1
constants.cm_unsteady	0.728
constants.sigma_rules	1
constants.cc_geometric	1
constants.cc_offset	0
constants.cc_floor	1
constants_source	results/calibrated_constants.json, read by unistall.dsmodel.load_constants
march_rule.largest_step_semichords	0.02
march_rule.settle_semichords	200
march_rule.steps_min	720
march_rule.cycles_min	3
march_rule.cycles_max	8
compared_range.k_max	0.2
compared_range.peak_alpha_max_deg	25
cases.A_validation.label	Case A: tunnel condition
cases.A_validation.description	tunnel condition of NASA TM-84245, frame 9302
cases.A_validation.alpha_mean_deg	10
cases.A_validation.alpha_amp_deg	10
cases.A_validation.reduced_frequency_k	0.1
cases.A_validation.mach_M	0.3

Key	Value
cases.A_validation.chord_m	0.61
cases.A_validation.measured_frame	frame_9302
cases.A_validation.march.steps_per_cycle	3240
cases.A_validation.march.cycles	5
cases.A_validation.march.step_semichords	0.019393
cases.A_validation.critical.sonic_speed_ms	313.4263
cases.A_validation.critical.Cp_critical	-6.9473
cases.A_validation.critical.Cp_vacuum	-15.873
cases.A_validation.static_inputs.CN_alpha_per_rad	6.70492
cases.A_validation.static_inputs.alpha0_deg	0.2343
cases.A_validation.static_inputs.CN_static_at_zero_incidence	-0.02741
cases.A_validation.static_inputs.alpha_stall_deg	13.6598
cases.A_validation.static_inputs.CN1_onset	1.34202
cases.A_validation.static_inputs.inside_static_mach_range	yes
cases.B_application.label	Case B: rotor-blade-station condition
cases.B_application.description	an aerofoil at a rotor-blade-station condition (steady stream, prescribed pitch); not a rotor calculation
cases.B_application.alpha_mean_deg	12
cases.B_application.alpha_amp_deg	8
cases.B_application.reduced_frequency_k	0.07471689346
cases.B_application.mach_M	0.28
cases.B_application.chord_m	0.527
cases.B_application.measured_frame	
cases.B_application.march.steps_per_cycle	4320
cases.B_application.march.cycles	4
cases.B_application.march.step_semichords	0.019466
cases.B_application.critical.sonic_speed_ms	313.0689
cases.B_application.critical.Cp_critical	-8.0567

Key	Value
cases.B_application.critical.Cp_vacuum	-18.2216
cases.B_application.static_inputs.CN_alpha_per_rad	6.71809
cases.B_application.static_inputs.alpha0_deg	0.1768
cases.B_application.static_inputs.CN_static_at_zero_incidence	-0.02073
cases.B_application.static_inputs.alpha_stall_deg	14.2575
cases.B_application.static_inputs.CN1_onset	1.40225
cases.B_application.static_inputs.inside_static_mach_range	yes
field_reconstruction.module	unistall.flowfield
field_reconstruction.what_it_is	Potential-flow reconstruction drawn round the predicted lift, not a flow solution: incompressible, with a linearised (Karman-Tsien) correction of the pressure that holds only where the local flow is subcritical, and quasi-steady (no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free); no boundary layer, no separated shear layer. A transpiration through the suction surface, of either sign, stands for the separated region, which is not drawn: no reversed flow and no wake appear. The vortex is a marker of the model's vortex clock and does not reproduce a suction footprint. Compared with no measurement; no load depends on it.
field_reconstruction.quasi_steady_omissions	no shed wake, no pitch-rate boundary condition, no unsteady pressure term, vortex not force-free
field_reconstruction.read_by_the_driver.note	04_solver/run_case.py passes these four to unistall.flowfield; changing one here changes the output
field_reconstruction.read_by_the_driver.n_panels	320
field_reconstruction.read_by_the_driver.domain_chords	-1, 2, -1.2, 1.2
field_reconstruction.read_by_the_driver.grid_nx	180
field_reconstruction.read_by_the_driver.grid_ny	140
field_reconstruction.record_of_module_constants.note	a record of the constants of unistall.flowfield at the time of writing; the driver does not read them, and a check asserts that they still equal the module's
field_reconstruction.record_of_module_constants.panels	straight panels, each with a source of constant strength and a vortex sheet of one uniform strength, velocities from the closed-form panel integrals
field_reconstruction.record_of_module_constants.near_wall_panel_lengths	0.6
field_reconstruction.record_of_module_constants.near_wall_rings_blanked	0
field_reconstruction.record_of_module_constants.attached_outflow_fraction	0.05
field_reconstruction.record_of_module_constants.vortex_x0	0.25

Key	Value
field_reconstruction.record_of_module_constants.vortex_dx	0.75
field_reconstruction.record_of_module_constants.vortex_y0	0.1
field_reconstruction.record_of_module_constants.vortex_dy	0.06
field_reconstruction.record_of_module_constants.vortex_core_over_c	0.05
field_reconstruction.record_of_module_constants.lamb_oseen_peak_coefficient	0.638173
field_reconstruction.transpiration.what	an equivalent transpiration through the suction surface standing for the separated region: an outflow where positive, an inflow where negative; its strength is solved so that the Kutta condition holds at the trailing edge while the circulation stays that of the predicted lift
field_reconstruction.transpiration.wall_condition	$v_n = v_t w$ on the suction-surface panels aft of $x/c = \min(f_{sep}, 1 - \text{attached_outflow_fraction})$; w a smooth step from 0 there to 1 at the trailing edge
field_reconstruction.transpiration.kutta_condition	equal pressure at the two control points next to the trailing edge
field_reconstruction.compressibility.what	a linearised correction applied to an incompressible reconstruction, valid only where the local flow is subcritical; not a compressible solution
field_reconstruction.compressibility.pressure	Karman-Tsien correction of the incompressible C_p at the stream Mach number: $C_{p_c} = C_p / (\beta + M^2 C_p / (2 (1 + \beta)))$, $\beta = \sqrt{1 - M^2}$
field_reconstruction.compressibility.vacuum_cap	0.95
field_reconstruction.compressibility.vacuum_cap_note	the corrected C_p is not allowed below this fraction of the vacuum value $-2/(\gamma M^2)$ (cases.<case>.critical. C_{p_vacuum})
field_reconstruction.compressibility.beyond_critical	where the corrected C_p is below cases.<case>.critical. $C_{p_critical}$ the local Mach number is above one, the correction does not hold and the values are not physical; those points are flagged, hatched in the figures and counted
field_reconstruction.compressibility.local_state	local Mach number, static temperature and recovery temperature from the corrected C_p by the isentropic relations with the stagnation state of the stream
field_reconstruction.compressibility.recovery_factor	0.8963
field_reconstruction.compressibility.recovery_factor_source	$Pr^{(1/3)}$, the usual value for a turbulent boundary layer on a flat plate, with $Pr = 0.72$ for air
field_reconstruction.compressibility.circulation	the sheet carries the fraction of the circulation of the lift at which the corrected surface pressure returns the lift given; the fraction is solved at each instant, starting from the Prandtl-Glauert factor $\sqrt{1 - M^2}$
field_reconstruction.field_files.format	gzip-compressed CSV, one row per grid node
field_reconstruction.field_files.columns	x_m , y_m , u_{ms} , v_{ms} , C_p , $C_{p_incompressible}$, $Mach$, T_{static_K} , $T_{recovery_K}$, $vorticity_{1s}$, $beyond_critical$, $inside$

Key	Value
field_reconstruction.field_files.velocity	u_ms, v_ms: the incompressible velocity
field_reconstruction.field_files.beyond_critical	1 where the corrected C_p is below the critical value: C_p , Mach and the temperatures are not physical there; 0 elsewhere
field_reconstruction.field_files.inside	1 at a node inside the section, where the values are those of the singularities and have no physical meaning; 0 elsewhere
field_reconstruction.bound_circulation	$\Gamma = 0.5 C_L U c$, uniform vortex sheet on the surface
field_reconstruction.vortex.what	a marker of the load model's vortex clock; it does not reproduce a suction footprint
field_reconstruction.vortex.circulation	$0.5 C_N \text{vortex } U c$
field_reconstruction.vortex.profile	Lamb-Oseen
field_reconstruction.vortex.position	$x/c = x_0 + dx \tau$, $y/c = y_0 + dy \tau$, $\tau = \tau_v / T_v$; over the trailing edge at $\tau = 1$
field_reconstruction.vortex.path_and_core	assumed: chosen for illustration, neither derived nor measured
field_reconstruction.field_phases.rise	upstroke, mean incidence plus half the amplitude
field_reconstruction.field_phases.peak	instant of greatest lift
field_reconstruction.field_phases.dsv	instant of greatest vortex normal force
field_reconstruction.field_phases.fall	downstroke, mean incidence plus half the amplitude

6 Calibrated constants and the record of the fit

Source: results/calibrated_constants.json (without the copy kept under first_stage_record). 82 rows.

Key	Value
author	Akosa Samuel Onyejekwe (independent)
model	tabulated
constants.Tp	1.651526
constants.Tf0	6.278239
constants.Tv0	6.317996
constants.Tvl	8.391549
constants.eta	0.906958
fixed.xcp_v	0.2
fixed.St_sh	0.19

Key	Value
fixed.cm_lag	0.1
fixed.CD0	0.0072
fixed.cn1_factor	1
fixed.cmq_kirchhoff	1
fixed.cm_unsteady	0.728
fixed.sigma_rules	1
fixed.cc_geometric	1
fixed.cc_offset	0
fixed.cc_floor	1
cost	1.90576495
cost_at_literature_values	2.14349292
cost_definition	sum over frames of $nRMS_CL^2 + nRMS_CM^2 + nRMS_CD^2$, at the reporting resolution
algorithm	scipy.optimize.least_squares, method trf, bounded; search at a coarser march, then every start refined at the reporting resolution
starts	8
seed	20261006
bounds.Tp	0.3, 4
bounds.Tf0	0.5, 10
bounds.Tv0	1, 15
bounds.Tvl	3, 13
bounds.eta	0.6, 1
march.search_step_semichords	0.32
march.search_settle_semichords	120
march.reporting_step_semichords	0.02
march.reporting_settle_semichords	200
march.jacobian_relative_step	0.01
frames	frame_7023, frame_7101, frame_7104, frame_7110, frame_7119, frame_7200, frame_7212, frame_7305, frame_9210, frame_9213, frame_9222, frame_9302, frame_10104, frame_10113, frame_10203, frame_10212, frame_10218, frame_10221, frame_10305, frame_14220

Key	Value
n_frames	20
spread_over_starts.Tp	1.2519, 2.033
spread_over_starts.Tf0	4.6053, 7.3605
spread_over_starts.Tv0	2.6486, 9.3007
spread_over_starts.Tvl	6.8408, 12.9876
spread_over_starts.eta	0.9063, 0.9094
spread_of_search_cost	1.90378, 2.01605
starts_within_1pct_of_best_search	3
polished_costs	1.918899, 1.962506, 2.013453, 1.907005, 1.959429, 1.905765, 2.01251, 1.907238
polished_constants_differ_by.Tp	0.8353
polished_constants_differ_by.Tf0	2.5325
polished_constants_differ_by.Tv0	5.9016
polished_constants_differ_by.Tvl	5.9277
polished_constants_differ_by.eta	0.0016
selection.method	5-fold cross-validation inside the calibration set
selection.chosen	all
selection.candidates.all.constants	Tp, Tf0, Tv0, Tvl, eta
selection.candidates.all.left_out_cost	1.978641
selection.candidates.sensitive only.constants	Tvl, eta
selection.candidates.sensitive only.left_out_cost	2.058586
selection.candidates.none.constants	
selection.candidates.none.left_out_cost	2.143493
selection.starts_per_fold	2
selection.rule	candidates in order of size: none, sensitive only, all. A larger set of fitted constants replaces the next smaller one only if its left-out cost is lower, averaged over the folds, by more than one standard error of the fold-by-fold differences
selection.steps.1.smaller	none
selection.steps.1.larger	sensitive only
selection.steps.1.mean_improvement_per_fold	0.016981

Key	Value
selection.steps.1.standard_error	0.009119
selection.steps.1.larger_kept	yes
selection.steps.2.smaller	sensitive only
selection.steps.2.larger	all
selection.steps.2.mean_improvement_per_fold	0.015989
selection.steps.2.standard_error	0.005143
selection.steps.2.larger_kept	yes
selection.folds_in_which_chosen_is_better.sensitive only	5
selection.folds_in_which_chosen_is_better.none	4
selection.sensitive_set_by_fold.0	Tvl eta
selection.sensitive_set_by_fold.1	Tp Tvl eta
selection.sensitive_set_by_fold.2	Tvl eta
selection.sensitive_set_by_fold.3	Tvl eta
selection.sensitive_set_by_fold.4	Tvl eta
selection.returned_to_literature_values	
selection.largest_cost_change_pct_for_10pct.Tp	0.256
selection.largest_cost_change_pct_for_10pct.Tf0	0.17
selection.largest_cost_change_pct_for_10pct.Tv0	0.083
selection.largest_cost_change_pct_for_10pct.Tvl	3.836
selection.largest_cost_change_pct_for_10pct.eta	30.928

7 Unsteady-moment factor (empirical; a correction of this model)

Source: results/attached_moment_factor.json. 151 rows.

Key	Value
author	Akosa Samuel Onyejekwe (independent)
cm_unsteady	0.728
bounds	0.3, 1.2
rule	the attached part of every calibration loop: measured moment points that the model, with literature stall constants, places before stall onset (see basis)
basis.cm_unsteady	0.728
basis.interval_95	0.606, 0.892
basis.n_loops	19
basis.n_points	184
basis.onset_margin_deg	1
basis.loops.1.frame	frame_7023
basis.loops.1.k	0.201
basis.loops.1.points	12
basis.loops.1.model_reaches_onset	yes
basis.loops.1.factor_this_loop	0.963
basis.loops.2.frame	frame_7101
basis.loops.2.k	0.15
basis.loops.2.points	10
basis.loops.2.model_reaches_onset	yes
basis.loops.2.factor_this_loop	1.004
basis.loops.3.frame	frame_7104
basis.loops.3.k	0.025
basis.loops.3.points	6
basis.loops.3.model_reaches_onset	yes
basis.loops.3.factor_this_loop	0.898
basis.loops.4.frame	frame_7110

Key	Value
basis.loops.4.k	0.1
basis.loops.4.points	8
basis.loops.4.model_reaches_onset	yes
basis.loops.4.factor_this_loop	0.477
basis.loops.5.frame	frame_7119
basis.loops.5.k	0.099
basis.loops.5.points	7
basis.loops.5.model_reaches_onset	yes
basis.loops.5.factor_this_loop	0.996
basis.loops.6.frame	frame_7200
basis.loops.6.k	0.025
basis.loops.6.points	5
basis.loops.6.model_reaches_onset	yes
basis.loops.6.factor_this_loop	-0.063
basis.loops.7.frame	frame_7212
basis.loops.7.k	0.149
basis.loops.7.points	9
basis.loops.7.model_reaches_onset	yes
basis.loops.7.factor_this_loop	0.59
basis.loops.8.frame	frame_7305
basis.loops.8.k	0.151
basis.loops.8.points	6
basis.loops.8.model_reaches_onset	yes
basis.loops.8.factor_this_loop	0.811
basis.loops.9.frame	frame_9210
basis.loops.9.k	0.01
basis.loops.9.points	7
basis.loops.9.model_reaches_onset	yes

Key	Value
basis.loops.9.factor_this_loop	3.342
basis.loops.10.frame	frame_9213
basis.loops.10.k	0.024
basis.loops.10.points	6
basis.loops.10.model_reaches_onset	yes
basis.loops.10.factor_this_loop	0.037
basis.loops.11.frame	frame_9222
basis.loops.11.k	0.024
basis.loops.11.points	8
basis.loops.11.model_reaches_onset	yes
basis.loops.11.factor_this_loop	0.502
basis.loops.12.frame	frame_9302
basis.loops.12.k	0.096
basis.loops.12.points	14
basis.loops.12.model_reaches_onset	yes
basis.loops.12.factor_this_loop	0.609
basis.loops.13.frame	frame_10104
basis.loops.13.k	0.048
basis.loops.13.points	6
basis.loops.13.model_reaches_onset	yes
basis.loops.13.factor_this_loop	0.646
basis.loops.14.frame	frame_10203
basis.loops.14.k	0.025
basis.loops.14.points	6
basis.loops.14.model_reaches_onset	yes
basis.loops.14.factor_this_loop	0.26
basis.loops.15.frame	frame_10212
basis.loops.15.k	0.198

Key	Value
basis.loops.15.points	10
basis.loops.15.model_reaches_onset	yes
basis.loops.15.factor_this_loop	0.904
basis.loops.16.frame	frame_10218
basis.loops.16.k	0.01
basis.loops.16.points	9
basis.loops.16.model_reaches_onset	no
basis.loops.16.factor_this_loop	1.406
basis.loops.17.frame	frame_10221
basis.loops.17.k	0.099
basis.loops.17.points	22
basis.loops.17.model_reaches_onset	no
basis.loops.17.factor_this_loop	0.598
basis.loops.18.frame	frame_10305
basis.loops.18.k	0.099
basis.loops.18.points	25
basis.loops.18.model_reaches_onset	yes
basis.loops.18.factor_this_loop	0.885
basis.loops.19.frame	frame_14220
basis.loops.19.k	0.103
basis.loops.19.points	8
basis.loops.19.model_reaches_onset	yes
basis.loops.19.factor_this_loop	0.859
factor_from_the_loops_below_static_stall_alone	0.647
range_over_loops_with_a_closed_moment_loop	0.598, 0.707
frames.1.frame	frame_7110
frames.1.M	0.301
frames.1.k	0.1

Key	Value
frames.1.moment_loop_closed	yes
frames.1.damping_measured	0.16
frames.1.damping_without_factor	0.2117
frames.1.damping_with_factor	0.1654
frames.1.damping_static_part	0.0417
frames.1.damping_unlagged_static	0.1699
frames.1.nRMS_CM_without_factor	0.1393
frames.1.nRMS_CM_with_factor	0.1182
frames.1.nRMS_CM_unlagged_static	0.088
frames.1.best_factor_this_loop	0.707
frames.1.peak_vortex_normal_force	0.0162
frames.1.peak_CN_prime_over_CN1	1.0545
frames.2.frame	frame_10218
frames.2.M	0.3
frames.2.k	0.01
frames.2.moment_loop_closed	no
frames.2.damping_measured	
frames.2.damping_without_factor	0.0216
frames.2.damping_with_factor	0.0169
frames.2.damping_static_part	0.0041
frames.2.damping_unlagged_static	0.0175
frames.2.nRMS_CM_without_factor	0.0976
frames.2.nRMS_CM_with_factor	0.1002
frames.2.nRMS_CM_unlagged_static	0.0973
frames.2.best_factor_this_loop	1.199
frames.2.peak_vortex_normal_force	0
frames.2.peak_CN_prime_over_CN1	0.8508
frames.3.frame	frame_10221

Key	Value
frames.3.M	0.301
frames.3.k	0.099
frames.3.moment_loop_closed	yes
frames.3.damping_measured	0.1395
frames.3.damping_without_factor	0.1998
frames.3.damping_with_factor	0.1535
frames.3.damping_static_part	0.0297
frames.3.damping_unlagged_static	0.1701
frames.3.nRMS_CM_without_factor	0.1705
frames.3.nRMS_CM_with_factor	0.1171
frames.3.nRMS_CM_unlagged_static	0.128
frames.3.best_factor_this_loop	0.598
frames.3.peak_vortex_normal_force	0
frames.3.peak_CN_prime_over_CN1	0.8008

8 Constants, literature and calibrated

Source: results/tables.md, Table 3. Constants. 6 rows.

Constant	Literature	Tabulated model	Fitted-law model	Status	Range over starts
pressure lag T_p	1.700	1.652	1.602	fitted; the two refined starts differ by 0.835	1.25 to 2.03
boundary-layer lag T_f	3.000	6.278	6.314	fitted; the two refined starts differ by 2.533	4.61 to 7.36
vortex decay T_v	6.000	6.318	7.269	fitted; the two refined starts differ by 5.902	2.65 to 9.30
vortex travel time T_{vl}	7.000	8.392	8.274	fitted; the two refined starts differ by 5.928	6.84 to 12.99
chord-force recovery η	0.950	0.907	0.909	fitted; the two refined starts differ by 0.002	0.91 to 0.91
unsteady-moment factor g_M	1.000	0.728	0.728	empirical, from loops below static stall	

9 Static inputs at the Mach stations

Source: results/static_by_mach_naca0012.csv. 2 rows.

M	frame	set	CL alpha per deg	CN alpha per rad	alpha zero lift deg	CL max	alpha at CL max deg	CN at CL max	fit range deg	source
0.215	frame_13308	calibration	0.118	6.759	-0.01	1.644	16.2	1.598	-5 to 8 (up sweep)	NASA TM-84245 quasi-steady sweep frame_13308 (k = 0.001), not wall-corrected, same measurement chain as the dynamic frames
0.302	frame_12102	calibration	0.117	6.703	0.24	1.364	13.6	1.336	-5 to 8 (up sweep)	NASA TM-84245 quasi-steady sweep frame_12102 (k = 0.001), not wall-corrected, same measurement chain as the dynamic frames

10 Static inputs of the cases, sampled (one row in 8 of 202)

Source: 03_model_setup/static_inputs.csv. 26 rows.

case	mach M	alpha deg	CN static	CM static c4	f separation
A_validation	0.3	0	-0.02741	-0.00529	1
A_validation	0.3	2	0.20663	-0.00441	1
A_validation	0.3	4	0.44061	-0.00182	0.99968
A_validation	0.3	6	0.65123	0.00387	0.93096
A_validation	0.3	8	0.88583	0.00846	0.94982
A_validation	0.3	10	1.09949	0.01325	0.92491
A_validation	0.3	12	1.27126	0.02267	0.84966
A_validation	0.3	14	1.34669	0.02686	0.68664
A_validation	0.3	16	0.92161	-0.09127	0.17102
A_validation	0.3	18	0.86001	-0.0792	0.08199
A_validation	0.3	20	0.90513	-0.09116	0.06305
A_validation	0.3	22	0.91798	-0.09354	0.04027
A_validation	0.3	24	1.02895	-0.09354	0.04688
B_application	0.28	0	-0.02073	-0.00471	1
B_application	0.28	2	0.21378	-0.00356	1

case	mach M	alpha deg	CN static	CM static c4	f separation
B_application	0.28	4	0.44783	-0.00095	0.99796
B_application	0.28	6	0.66377	0.00436	0.94468
B_application	0.28	8	0.89227	0.0092	0.94583
B_application	0.28	10	1.11054	0.01385	0.929
B_application	0.28	12	1.27929	0.02146	0.84871
B_application	0.28	14	1.40992	0.02978	0.74884
B_application	0.28	16	0.95397	-0.07193	0.18847
B_application	0.28	18	0.85961	-0.07961	0.07992
B_application	0.28	20	0.8888	-0.08272	0.05605
B_application	0.28	22	0.91134	-0.09395	0.03747
B_application	0.28	24	0.9912	-0.09395	0.03662

11 Section geometry

Source: 01_geometry/section_geometry_summary.csv. 8 rows.

property	value	units
section	NACA 0012	-
trailing_edge	closed: x^4 coefficient -0.1036 of the four-digit thickness form	-
max_thickness_t_c	0.12001	-
x_at_max_thickness_x_c	0.2995	-
LE_radius_r_c	0.01587	-
TE_gap_percent_chord	0.0	%
enclosed_area_A_c2	0.0817	-
n_surface_points	319	-

12 Mesh quality (the load model does not read the grid)

Source: 02_mesh/mesh_quality_metrics.csv. 26 rows.

metric	value	unit
topology	O-grid, no wake cut, seam at the trailing edge	-
status	not a CFD mesh that has been run; not read by the load model	-
nodes_wrap_i	193	-
nodes_normal_j	81	-
total_nodes	15633	-
unique_nodes	15552	-
total_cells	15360	-
farfield_radius	20.0	chord
farfield_radius_deviation_max	7.1×10^{-15}	chord
first_layer_height	0.001	chord
wall_normal_growth_ratio	1.0996	-
wall_spacing_min	0.000457	chord
wall_spacing_max	0.0183	chord
wall_spacing_at_leading_edge	0.00203	chord
wall_spacing_at_trailing_edge	0.000457	chord
max_skewness_equiangle	0.908	-
max_skewness_cell_i_j	0 0	-
max_skewness_without_trailing_edge_cells	0.491	-
mean_skewness_equiangle	0.063	-
cells_with_skewness_above_0.5	0.01	percent
min_orthogonality_angle	8.3	deg
min_orthogonality_angle_without_trailing_edge_cells	45.8	deg
max_aspect_ratio	18.3	-
mean_aspect_ratio	3.9	-
min_cell_area_signed	5.131×10^{-7}	chord ²
inverted_cells	0	-

13 Metrics of the cases (cycle damping is model output only)

Source: 05_solution/metrics_A_validation.csv and 05_solution/metrics_B_application.csv. 147 rows.

metric	A validation	B application	units
CL_max	1.79505	1.74544	-
alpha_at_CLmax_deg	18.0673	17.0255	deg
CM_min_c4	-0.289612	-0.236861	-
alpha_at_CMmin_deg	19.6285	18.0605	deg
CD_max	0.555432	0.4361	-
alpha_at_CDmax_deg	19.6285	18.0605	deg
CD_min	0.00370718	0.00910269	-
stall_onset_alpha_deg	14.4267	13.8466	deg
vortex_sheddings_per_cycle	2	3	-
vortex_clock_runs_per_cycle	2	3	-
cycle_damping_Xi	0.0889893	0.0646719	-
Cp_closure_error_pct	1.0608×10^{-5}	2.5569×10^{-6}	%
Cp_closure_dCL	1.9041×10^{-7}	4.4629×10^{-8}	-
circulation_factor_at_CLmax	0.89289	0.88875	-
prandtl_glauert_factor	0.95394	0.96	-
Cp_closure_error_pct_prandtl_glauert_factor	6.1011	7.5203	%
Cp_closure_error_pct_incompressible	-0.9962	-0.66754	%
Cp_closure_error_pct_panels_only	-0.72979	-0.65984	%
Cp_closure_worst_abs_pct_stored_instants	2.8123×10^{-5}	2.5569×10^{-6}	%
Cp_closure_worst_abs_pct_incompressible_stored_instants	3.4524	3.5496	%
Cp_closure_worst_abs_dCL_over_cycle	9.72402×10^{-7}	9.77568×10^{-7}	-
circulation_factor_cycle_min	0.880732	0.884284	-
circulation_factor_cycle_max	0.963048	0.966546	-
Cp_moment_minus_model_at_CLmax	0.27744	0.24374	-

metric	A validation	B application	units
Cp_drag_minus_model_at_CLmax	-0.46637	-0.41462	-
Cp_moment_minus_model_worst_abs_stored_instants	0.41492	0.3628	-
Cp_drag_minus_model_worst_abs_stored_instants	0.68229	0.59357	-
Cp_moment_sign_agrees_with_model_instants	2	3	-
stored_instants	6	6	-
Cp_trailing_edge_jump_worst_stored_instants	1.1382×10^{-8}	6.4851×10^{-9}	-
Cp_critical	-6.94732	-8.05673	-
Cp_vacuum	-15.873	-18.2216	-
sonic_speed_over_U	3.07017	3.28571	-
total_temperature_K	293.337	292.668	K
recovery_factor	0.896281	0.896281	-
Cp_surface_min_stored_instants	-15.079	-17.31	-
Cp_surface_min_incompressible_stored_instants	-16	-14.659	-
beyond_critical_surface_fraction_worst_stored_instants	0.032054	0.022925	-
beyond_critical_surface_fraction_worst_over_cycle	0.0330586	0.0248481	-
beyond_critical_fraction_of_cycle	0.373961	0.373961	-
beyond_critical_field_area_over_c2_worst_stored_instants	0.00086813	0.00057875	-
beyond_critical_field_fraction_worst_stored_instants	0.00012034	8.0225×10^{-5}	-
cycle_instants_scanned	361	361	-
transpiration_vt_over_U_cycle_min	-0.0593495	0.0324499	-
transpiration_vt_over_U_cycle_max	0.597222	0.582637	-
transpiration_inflow_fraction_of_cycle	0.204986	0	-
transpiration_flux_over_Uc_cycle_min	-0.00770406	0.00324785	-
transpiration_flux_over_Uc_cycle_max	0.259151	0.268318	-
transpiration_vt_over_U_rise	0.42071	0.24989	-
transpiration_flux_over_Uc_rise	0.018543	0.02205	-
transpiration_x_sep_rise	0.91268	0.82511	-
circulation_total_over_Uc_rise	0.69796	0.79908	-

metric	A validation	B application	units
circulation_factor_rise	0.89151	0.90095	-
Cp_surface_min_rise	-14.395	-17.157	-
Cp_surface_max_rise	1.0235	1.0204	-
Cp_field_min_subcritical_rise	-6.9443	-7.7974	-
Cp_field_max_rise	1.0091	1.0179	-
Mach_surface_max_subcritical_rise	0.99072	0.96869	-
Mach_field_max_subcritical_rise	0.99971	0.97825	-
T_static_surface_min_subcritical_K_rise	245.2	246.42	K
T_static_field_min_subcritical_K_rise	244.47	245.65	K
T_recovery_surface_max_K_rise	293.34	292.67	K
beyond_critical_surface_fraction_rise	0.019556	0.020276	-
beyond_critical_field_area_over_c2_rise	0	0	-
beyond_critical_field_fraction_rise	0	0	-
closure_error_pct_rise	2.9457×10^{-8}	-5.1572×10^{-7}	-
closure_error_pct_incompressible_rise	-0.7276	-1.0923	-
dCp_under_vortex_rise	0.072378	0.24862	-
transpiration_vt_over_U_peak	0.21057	0.22395	-
transpiration_flux_over_Uc_peak	0.038329	0.038088	-
transpiration_x_sep_peak	0.63888	0.66263	-
circulation_total_over_Uc_peak	0.92225	0.87388	-
circulation_factor_peak	0.89289	0.88875	-
Cp_surface_min_peak	-15.079	-17.31	-
Cp_surface_max_peak	1.0236	1.02	-
Cp_field_min_subcritical_peak	-6.1074	-6.1321	-
Cp_field_max_peak	1.0235	1.0163	-
Mach_surface_max_subcritical_peak	0.98534	0.97979	-
Mach_field_max_subcritical_peak	0.92076	0.84261	-
T_static_surface_min_subcritical_K_peak	245.64	245.53	K

metric	A validation	B application	units
T_static_field_min_subcritical_K_peak	250.81	256.28	K
T_recovery_surface_max_K_peak	293.34	292.67	K
beyond_critical_surface_fraction_peak	0.032054	0.022925	-
beyond_critical_field_area_over_c2_peak	0.00086813	0.00057875	-
beyond_critical_field_fraction_peak	0.00012034	8.0225×10^{-5}	-
closure_error_pct_peak	1.0608×10^{-5}	2.5569×10^{-6}	-
closure_error_pct_incompressible_peak	-0.9962	-0.66754	-
dCp_under_vortex_peak	0.55838	0.35856	-
transpiration_vt_over_U_dsv	0.54412	0.45283	-
transpiration_flux_over_Uc_dsv	0.19492	0.13335	-
transpiration_x_sep_dsv	0.28809	0.41513	-
circulation_total_over_Uc_dsv	1.0153	0.95617	-
circulation_factor_dsv	0.89795	0.89306	-
Cp_surface_min_dsv	-15.079	-17.31	-
Cp_surface_max_dsv	1.0232	1.0204	-
Cp_field_min_subcritical_dsv	-6.0232	-6.5603	-
Cp_field_max_dsv	1.0187	1.0203	-
Mach_surface_max_subcritical_dsv	0.99002	0.99467	-
Mach_field_max_subcritical_dsv	0.91294	0.87699	-
T_static_surface_min_subcritical_K_dsv	245.26	244.32	K
T_static_field_min_subcritical_K_dsv	251.43	253.65	K
T_recovery_surface_max_K_dsv	293.34	292.67	K
beyond_critical_surface_fraction_dsv	0.030873	0.022925	-
beyond_critical_field_area_over_c2_dsv	0.00086813	0.00057875	-
beyond_critical_field_fraction_dsv	0.00012034	8.0225×10^{-5}	-
closure_error_pct_dsv	2.8123×10^{-5}	-1.8537×10^{-6}	-
closure_error_pct_incompressible_dsv	-1.5808	-1.3954	-
dCp_under_vortex_dsv	-0.13684	-0.10863	-

metric	A validation	B application	units
transpiration_vt_over_U_fall	0.44454	0.46371	-
transpiration_flux_over_Uc_fall	0.20394	0.21482	-
transpiration_x_sep_fall	0.087517	0.078554	-
circulation_total_over_Uc_fall	0.30486	0.33537	-
circulation_factor_fall	0.93613	0.94536	-
Cp_surface_min_fall	-5.8833	-6.6332	-
Cp_surface_max_fall	1.0235	1.0204	-
Cp_field_min_subcritical_fall	-3.5542	-3.8362	-
Cp_field_max_fall	1.0232	1.0097	-
Mach_surface_max_subcritical_fall	0.90001	0.88288	-
Mach_field_max_subcritical_fall	0.68726	0.65822	-
T_static_surface_min_subcritical_K_fall	252.44	253.2	K
T_static_field_min_subcritical_K_fall	268.02	269.33	K
T_recovery_surface_max_K_fall	293.34	292.67	K
beyond_critical_surface_fraction_fall	0	0	-
beyond_critical_field_area_over_c2_fall	0	0	-
beyond_critical_field_fraction_fall	0	0	-
closure_error_pct_fall	4.3178×10^{-8}	2.9867×10^{-8}	-
closure_error_pct_incompressible_fall	-2.6023	-2.6929	-
dCp_under_vortex_fall			-
CN_vortex_max	0.513113	0.390203	-
alpha_at_CN_vortex_max_deg	19.6285	18.0605	deg
vortex_clock_tau_over_Tvl	0.999567	0.999449	-
vortex_circulation_over_Uc	0.256556	0.195102	-
vortex_peak_swirl_over_U	0.52116	0.396323	-
vortex_peak_vorticity_c_over_U	32.6658	24.8411	-
vortex_centre_Cp	-0.994139	-0.559596	-
vortex_x_over_c	0.999675	0.999587	- (assumed path)

metric	A validation	B application	units
vortex_y_over_c	0.159974	0.159967	- (assumed path)
vortex_core_radius_over_c	0.05	0.05	- (assumed)
mean_alpha_deg	10	12	deg
amp_alpha_deg	10	8	deg
reduced_frequency_k	0.1	0.0747169	-
mach_M	0.3	0.28	-
chord_m	0.61	0.527	m
freestream_velocity_U	102.088	95.2818	m/s
reynolds_number_Re_c	4263000	3438000	- (sea-level standard air)
steps_per_cycle	3240	4320	-
cycles_marched	5	4	-

14 Summary of the cases

Source: 05_solution/summary_all_cases.csv. 12 rows.

quantity	A validation	B application
CL_max	1.79505	1.74544
alpha_at_CLmax_deg	18.0673	17.0255
CM_min_c4	-0.289612	-0.236861
CD_max	0.555432	0.4361
stall_onset_alpha_deg	14.4267	13.8466
cycle_damping_Xi	0.0889893	0.0646719
Cp_closure_error_pct	1.0608×10^{-5}	2.5569×10^{-6}
Cp_closure_error_pct_incompressible	-0.9962	-0.66754
Cp_closure_error_pct_panels_only	-0.72979	-0.65984
circulation_factor_at_CLmax	0.89289	0.88875
CN_vortex_max	0.513113	0.390203
vortex_circulation_over_Uc	0.256556	0.195102

15 Convergence with cycles marched, Case A: tunnel condition

Source: 05_solution/convergence/residuals_A_validation.csv. 8 rows.

cycles marched	CL max	CM min	CD max	max abs change CL	max abs change CM	max abs change CD	reported
1	1.731298963	-0.2536049418	0.451730618				no
2	1.796634484	-0.2898856033	0.5557275646	0.4967284959	0.1726397412	0.1525550224	no
3	1.795048575	-0.2896115307	0.5554321303	0.0024840114	0.0002759662	0.0002954343	no
4	1.795048501	-0.2896115154	0.5554321151	1.399×10^{-7}	3.12×10^{-8}	1.68×10^{-8}	no
5	1.795048501	-0.2896115154	0.5554321151	0	0	0	yes
6	1.795048501	-0.2896115154	0.5554321151	0	0	0	no
7	1.795048501	-0.2896115154	0.5554321151	0	0	0	no
8	1.795048501	-0.2896115154	0.5554321151	0	0	0	no

16 Convergence with cycles marched, Case B: rotor-blade-station condition

Source: 05_solution/convergence/residuals_B_application.csv. 8 rows.

cycles marched	CL max	CM min	CD max	max abs change CL	max abs change CM	max abs change CD	reported
1	1.70377411	-0.1782857504	0.3541785421				no
2	1.745844526	-0.2369517758	0.4361698319	0.4278322617	0.1315383006	0.0951219964	no
3	1.745437269	-0.2368611932	0.4360999444	0.0008370189	9.36363×10^{-5}	6.98875×10^{-5}	no
4	1.745437269	-0.2368611931	0.4360999443	5×10^{-10}	1×10^{-10}	0	yes
5	1.745437269	-0.2368611931	0.4360999443	0	0	0	no
6	1.745437269	-0.2368611931	0.4360999443	0	0	0	no
7	1.745437269	-0.2368611931	0.4360999443	0	0	0	no
8	1.745437269	-0.2368611931	0.4360999443	0	0	0	no

17 Convergence with the time step

Source: 05_solution/convergence/timestep_refinement.csv. 12 rows.

case	steps per cycle	step semichords	CL max	CM min	CD max	stall onset alpha deg	cycle damping Xi	reported	pct from finest CL max	pct from finest CM min	pct from finest CD max
A_validation	405	0.15514038	1.79600745	-0.28881674	0.55441482	14.42358363	0.0888058	no	0.05767245	0.25882239	0.17068997
A_validation	810	0.07757019	1.79550257	-0.28948606	0.55538326	14.42549957	0.08891825	no	0.02954478	0.02767515	0.00368967
A_validation	1620	0.03878509	1.79520927	-0.28943189	0.55520226	14.42629393	0.08891879	no	0.01320495	0.04638294	0.02890079
A_validation	3240	0.01939255	1.7950485	-0.28961152	0.55543212	14.42665873	0.08898929	yes	0.00424803	0.01564874	0.01248686
A_validation	6480	0.00969627	1.79503131	-0.28957938	0.55539028	14.42683304	0.08896375	no	0.00329026	0.00455216	0.00495371
A_validation	12960	0.00484814	1.79497225	-0.2895662	0.55536277	14.42691758	0.08895698	no	0	0	0
B_application	540	0.15572821	1.74583551	-0.23295509	0.42966207	13.84603436	0.06214417	no	0.02481115	1.6423748	1.46928631
B_application	1080	0.07786411	1.74562023	-0.2352452	0.43340118	13.84647754	0.0636118	no	0.01247745	0.67545155	0.61182887
B_application	2160	0.03893205	1.74546751	-0.23634815	0.43522285	13.84659661	0.06433402	no	0.00372747	0.209767	0.19408081
B_application	4320	0.01946603	1.74543727	-0.23686119	0.43609994	13.8466305	0.06467194	yes	0.00199474	0.00684856	0.00705605
B_application	8640	0.00973301	1.7454225	-0.23684569	0.43607565	13.84664104	0.06467235	no	0.00114884	0.00030256	0.0014849
B_application	17280	0.00486651	1.74540245	-0.23684497	0.43606918	13.8466447	0.06467888	no	0	0	0

18 Observed order of convergence and discretisation uncertainty

Source: 05_solution/convergence/timestep_order.csv. 8 rows.

case	quantity	steps reported	value reported	value twice the steps	value four times the steps	monotone	observed order	order observed	richardson estimate	discretisation uncertainty	uncertainty pct of reported	uncertainty basis
A_validation	CL_max	3240	1.7950485	1.79503131	1.79497225	yes	-1.7804579	no		7.625×10^{-5}	0.00424785	range of the three finest marches (no order observed)
A_validation	CM_min	3240	-0.28961152	-0.28957938	-0.2895662	yes	1.28549283	yes	-0.28955703	5.448×10^{-5}	0.01881217	distance of the reported value from the Richardson estimate

case	quantity	steps reported	value reported	value twice the steps	value four times the steps	monotone	observed order	order observed	richardson estimate	discretisation uncertainty	uncertainty pct of reported	uncertainty basis
A_validation	CD_max	3240	0.55543212	0.55539028	0.55536277	yes	0.60474225	yes	0.55530993	0.00012218	0.02199755	distance of the reported value from the Richardson estimate
A_validation	stall_onset_alpha_deg	3240	14.42665873	14.42683304	14.42691758	yes	1.04374268	yes	14.42699723	0.0003385	0.00234633	distance of the reported value from the Richardson estimate
B_application	CL_max	4320	1.74543727	1.7454225	1.74540245	yes	-0.44160806	no		3.482×10^{-5}	0.0019947	range of the three finest marches (no order observed)
B_application	CM_min	4320	-0.23686119	-0.23684569	-0.23684497	yes	4.43530588	no		1.622×10^{-5}	0.00684809	range of the three finest marches (no order observed)
B_application	CD_max	4320	0.43609994	0.43607565	0.43606918	yes	1.90761092	yes	0.43606682	3.312×10^{-5}	0.00759511	distance of the reported value from the Richardson estimate
B_application	stall_onset_alpha_deg	4320	13.8466305	13.84664104	13.8466447	yes	1.52353728	yes	13.84664666	1.615×10^{-5}	0.00011666	distance of the reported value from the Richardson estimate

19 The reconstruction against the number of panels: largest change on doubling

Source: 05_solution/convergence/panel_convergence_summary.csv. 8 rows.

case	quantity	change measured as	n panels configured	largest change half to configured	largest change configured to double	tolerance	within tolerance
A_validation	Cp_surface_min_incompressible	percent of the value	320	0.9405	0.3675	1	yes
A_validation	transpiration_vt_over_U	absolute	320	0.001438	0.000991	0.005	yes
A_validation	closure_error_pct_incompressible	percentage points	320	0.3727	0.1903	0.5	yes
A_validation	beyond_critical_surface_fraction	absolute	320	0.002544	0.001085	0.005	yes
B_application	Cp_surface_min_incompressible	percent of the value	320	1.149	0.3423	1	yes
B_application	transpiration_vt_over_U	absolute	320	0.001001	0.000774	0.005	yes
B_application	closure_error_pct_incompressible	percentage points	320	0.3244	0.1663	0.5	yes
B_application	beyond_critical_surface_fraction	absolute	320	0.001754	0.001397	0.005	yes

20 The reconstruction against the number of panels, Case A: tunnel condition

Source: 05_solution/convergence/panel_convergence_A_validation.csv. 24 rows.

phase	n panels	configured	Cp surface min incompressible	transpiration vt over U	transpiration flux over Uc	circulation factor	beyond critical surface fraction	closure error pct	closure error pct incompressible	closure error pct panels only	Cp trailing edge jump	CM from Cp	CD from Cp	steepest panel aft of 0p98 deg
mean_up	80	no	-4.98723	0.332698	0.00963876	0.93496	0	-9.36252×10^{-5}	-1.5884	-2.50515	4.05231×10^{-15}	0.0511961	-0.0208644	8.26178
mean_up	160	no	-5.01136	0.333608	0.00970767	0.927079	0	-6.4209×10^{-5}	-0.867923	-1.40151	1.73472×10^{-15}	0.0509479	-0.0196201	8.26793
mean_up	320	yes	-5.05894	0.334812	0.00975419	0.922995	0	-3.31269×10^{-7}	-0.495266	-0.76382	1.24276×10^{-14}	0.0508285	-0.0189365	8.2697
mean_up	640	no	-5.07122	0.335725	0.0097851	0.920909	0	4.85747×10^{-5}	-0.304972	-0.408144	1.14093×10^{-13}	0.0507669	-0.0185715	8.27019
rise	80	no	-10.2465	0.418162	0.0183116	0.901238	0.0220604	3.71675×10^{-7}	-1.62754	-2.20594	4.14183×10^{-14}	0.0830856	-0.0786671	8.26178
rise	160	no	-10.2479	0.419271	0.0184634	0.894848	0.0220996	4.69682×10^{-8}	-1.03655	-1.22453	8.17402×10^{-15}	0.0829021	-0.0743851	8.26793
rise	320	yes	-10.3127	0.420709	0.0185427	0.891514	0.0195555	2.94572×10^{-8}	-0.727599	-0.661536	2.33424×10^{-14}	0.0828245	-0.0721848	8.2697
rise	640	no	-10.3165	0.4217	0.0185908	0.889815	0.0204636	1.97232×10^{-8}	-0.569338	-0.34952	1.70974×10^{-14}	0.0827822	-0.071037	8.27019

phase	n panels	configured	Cp surface min incompressible	transpiration vt over U	transpiration flux over Uc	circulation factor	beyond critical surface fraction	closure error pct	closure error pct incompressible	closure error pct panels only	Cp trailing edge jump	CM from Cp	CD from Cp	steepest panel aft of 0p98 deg
peak	80	no	-15.6287	0.213333	0.0388009	0.898023	0.0340889	1.09824×10^{-5}	-1.57778	-2.42772	3.50706×10^{-10}	0.13537	-0.128467	8.26178
peak	160	no	-15.418	0.21146	0.0384826	0.894582	0.0320415	-1.96258×10^{-5}	-1.20135	-1.3529	8.75145×10^{-11}	0.137459	-0.128596	8.26793
peak	320	yes	-15.3945	0.210574	0.0383286	0.892886	0.0320538	1.06076×10^{-5}	-0.996195	-0.729792	2.18557×10^{-11}	0.138377	-0.128223	8.2697
peak	640	no	-15.3801	0.210455	0.0383085	0.891736	0.0314441	7.98983×10^{-6}	-0.88921	-0.381734	5.48439×10^{-12}	0.139001	-0.127908	8.27019
dsv	80	no	-16.4702	0.543829	0.194779	0.90413	0.0340889	2.37187×10^{-5}	-2.18511	-3.62198	1.80087×10^{-7}	0.123708	-0.124671	8.26178
dsv	160	no	-16.0496	0.544416	0.19502	0.899293	0.0320415	-1.58865×10^{-8}	-1.79061	-2.03397	4.52741×10^{-8}	0.124997	-0.127866	8.26793
dsv	320	yes	-16.0003	0.544116	0.19492	0.897954	0.0308732	2.81233×10^{-5}	-1.58083	-1.08872	1.13822×10^{-8}	0.12531	-0.126863	8.2697
dsv	640	no	-15.9954	0.544418	0.195031	0.8967	0.0319584	1.24669×10^{-5}	-1.47301	-0.550584	2.86083×10^{-9}	0.125645	-0.126836	8.27019
fall	80	no	-4.81014	0.446997	0.205061	0.933339	0	8.58048×10^{-8}	-2.35232	-13.9711	1.44329×10^{-15}	0.0835822	-0.0025574	8.26178
fall	160	no	-4.89691	0.445298	0.204291	0.935345	0	5.37379×10^{-8}	-2.53253	-7.94968	9.54792×10^{-14}	0.0854323	-0.0031846	8.26793
fall	320	yes	-4.94264	0.444536	0.203944	0.936131	0	4.31776×10^{-8}	-2.60225	-4.23704	1.05249×10^{-13}	0.0863328	-0.00342325	8.2697
fall	640	no	-4.9558	0.444176	0.20378	0.93647	0	3.88682×10^{-8}	-2.6318	-2.07015	3.12972×10^{-13}	0.0867785	-0.00351565	8.27019
mean_down	80	no	-3.15073	0.248648	0.0997092	0.963159	0	5.04291×10^{-6}	-3.61076	-9.08062	9.5271×10^{-15}	0.0611332	-0.00355043	8.26178
mean_down	160	no	-3.25459	0.247259	0.0991637	0.96224	0	-2.56732×10^{-7}	-3.52269	-5.26193	2.00812×10^{-14}	0.0625195	-0.00403132	8.26793
mean_down	320	yes	-3.27744	0.246651	0.0989225	0.961533	0	-2.02614×10^{-6}	-3.45242	-2.92367	3.92048×10^{-14}	0.0631796	-0.00421431	8.2697
mean_down	640	no	-3.28953	0.246372	0.0988114	0.961111	0	-2.6104×10^{-6}	-3.40994	-1.56116	3.04756×10^{-14}	0.0635001	-0.00428727	8.27019

21 The reconstruction against the number of panels, Case B: rotor-blade-station condition

Source: 05_solution/convergence/panel_convergence_B_application.csv. 24 rows.

phase	n panels	configured	Cp surface min incompressible	transpiration vt over U	transpiration flux over U _c	circulation factor	beyond critical surface fraction	closure error pct	closure error pct incompressible	closure error pct panels only	Cp trailing edge jump	CM from Cp	CD from Cp	steepest panel aft of 0p98 deg
mean_up	80	no	-7.09719	0.303245	0.0100657	0.934426	0.00611371	3.30224×10^{-8}	-1.42721	-2.00574	6.91114×10^{-15}	0.0486965	-0.0324853	8.26178
mean_up	160	no	-7.24298	0.303563	0.0101565	0.927726	0.00517055	2.47549×10^{-8}	-0.806323	-1.10624	1.04777×10^{-14}	0.0485365	-0.0304578	8.26793
mean_up	320	yes	-7.25073	0.304564	0.0102014	0.924223	0.00517494	1.50087×10^{-8}	-0.48195	-0.595878	1.6237×10^{-15}	0.0484622	-0.02936	8.2697
mean_up	640	no	-7.26457	0.305338	0.0102305	0.92243	0.00603743	9.67345×10^{-9}	-0.315692	-0.315491	1.05055×10^{-14}	0.0484219	-0.0287747	8.27019
rise	80	no	-12.2826	0.25108	0.0221001	0.907752	0.0220604	-8.68406×10^{-7}	-1.78912	-1.94025	1.1497×10^{-11}	0.0843028	-0.110773	8.26178
rise	160	no	-12.2417	0.250051	0.022053	0.903322	0.0185222	-5.56929×10^{-7}	-1.33549	-1.06839	2.87748×10^{-12}	0.0852758	-0.106729	8.26793
rise	320	yes	-12.2627	0.249891	0.0220499	0.900953	0.0202759	-5.15715×10^{-7}	-1.09229	-0.573656	7.03854×10^{-13}	0.0857456	-0.104632	8.2697
rise	640	no	-12.2671	0.249917	0.0220549	0.899732	0.0202784	-4.7209×10^{-7}	-0.966267	-0.301925	8.36831×10^{-14}	0.085972	-0.103524	8.27019
peak	80	no	-13.9728	0.224134	0.0380767	0.896839	0.0220604	3.99121×10^{-6}	-1.37736	-2.20552	8.35208×10^{-10}	0.123932	-0.146035	8.26178
peak	160	no	-13.8591	0.224053	0.0380981	0.891307	0.0220996	4.23426×10^{-6}	-0.915493	-1.22393	2.08471×10^{-10}	0.12645	-0.146685	8.26793
peak	320	yes	-13.8495	0.223954	0.0380881	0.88875	0.0229252	2.55689×10^{-6}	-0.667538	-0.659841	4.97405×10^{-11}	0.127542	-0.145537	8.2697
peak	640	no	-13.8391	0.223846	0.0380717	0.887576	0.022928	5.52244×10^{-7}	-0.539065	-0.346858	1.23895×10^{-11}	0.128008	-0.144571	8.27019
dsv	80	no	-14.9254	0.451643	0.13295	0.901575	0.0220604	8.94388×10^{-6}	-2.08929	-2.88627	1.031×10^{-7}	0.123023	-0.156398	8.26178
dsv	160	no	-14.6955	0.452402	0.133213	0.895781	0.0238238	1.111×10^{-5}	-1.6369	-1.61843	2.58566×10^{-8}	0.125055	-0.157811	8.26793
dsv	320	yes	-14.6591	0.452832	0.133349	0.893059	0.0229252	-1.8537×10^{-6}	-1.39539	-0.874366	6.48505×10^{-9}	0.125941	-0.157471	8.2697
dsv	640	no	-14.642	0.453076	0.133423	0.891736	0.0243218	-2.94355×10^{-6}	-1.27084	-0.454848	1.62606×10^{-9}	0.126356	-0.156972	8.27019
fall	80	no	-5.45784	0.466119	0.215929	0.942996	0	5.8787×10^{-8}	-2.47991	-13.2622	3.96905×10^{-14}	0.0828753	-0.00149484	8.26178
fall	160	no	-5.59026	0.464456	0.215167	0.944705	0	3.69362×10^{-8}	-2.6344	-7.50613	3.81917×10^{-14}	0.0846952	-0.00202463	8.26793

phase	n panels	configured	Cp surface min incompressible	transpiration vt over U	transpiration flux over Uc	circulation factor	beyond critical surface fraction	closure error pct	closure error pct incompressible	closure error pct panels only	Cp trailing edge jump	CM from Cp	CD from Cp	steepest panel aft of 0p98 deg
fall	320	yes	-5.62206	0.463707	0.214823	0.945358	0	2.98672×10^{-8}	-2.69292	-3.96009	1.45661×10^{-13}	0.085584	-0.00220772	8.2697
fall	640	no	-5.63762	0.463352	0.214659	0.945635	0	2.7×10^{-8}	-2.71741	-1.89312	3.72036×10^{-13}	0.0860254	-0.00226946	8.27019
mean_down	80	no	-4.33717	0.292487	0.119737	0.968025	0	9.42002×10^{-6}	-3.70666	-8.8008	6.71685×10^{-15}	0.0679744	-0.00551422	8.26178
mean_down	160	no	-4.39233	0.291023	0.119149	0.967049	0	3.64133×10^{-6}	-3.61792	-5.07328	2.45498×10^{-14}	0.0694981	-0.00585581	8.26793
mean_down	320	yes	-4.4434	0.290384	0.11889	0.966331	0	1.46323×10^{-6}	-3.54956	-2.7922	3.21895×10^{-14}	0.0702245	-0.00595377	8.2697
mean_down	640	no	-4.45866	0.29009	0.118771	0.96591	0	6.3163×10^{-7}	-3.50874	-1.46458	4.85723×10^{-15}	0.0705781	-0.00597749	8.27019

22 CPU time of the reconstruction

Source: 05_solution/convergence/reconstruction_timing.csv. 2 rows.

case	what	instants	fields	cycle	instants scanned	n panels	cpu time s
A_validation	the six instants, the four fields and the scan of the cycle	6	4	361		320	11.68
B_application	the six instants, the four fields and the scan of the cycle	6	4	361		320	11.99

23 Error of the load model in each headline quantity on the held-out loops

Source: 05_solution/held_out_error_bands.csv. 5 rows.

quantity	error is	n loops	mean error	std error	mean abs error	largest abs error	source
CL_max	model minus measured, percent of measured	45	-3.3392	5.3378	4.5306	19.5431	results/validation_table.csv: CLmax_model, CLmax_exp
CM_min_c4	model minus measured, in the units of the quantity	45	0.0167	0.0508	0.0395	0.144	results/validation_table.csv: CMmin_model, CMmin_exp
CD_max	model minus measured, in the units of the quantity	45	-0.0525	0.0765	0.069	0.2408	results/validation_table.csv: CDmax_model, CDmax_exp
alpha_at_CLmax_deg	model minus measured, in the units of the quantity	45	-0.2677	0.8138	0.5723	3.7518	results/validation_table.csv: alpha_CLmax_model, alpha_CLmax_exp

quantity	error is	n loops	mean error	std error	mean abs error	largest abs error	source
moment_stall_alpha_deg	model minus measured, in the units of the quantity	29	-0.6083	0.6898	0.6216	2.671	results/validation_table.csv: alpha_Mstall_model, alpha_Mstall_exp

24 Case B with the static inputs of each Mach station

Source: 05_solution/static_station_spread.csv. 4 rows.

case	static inputs	static mach	CL max	CM min	CD max	stall onset alpha deg
B_application	interpolated to the case's Mach number (reported)	0.28	1.74544	-0.23686	0.4361	13.84663
B_application	Mach station 0.215	0.215	1.907	-0.29343	0.5182	15.29105
B_application	Mach station 0.302	0.302	1.70216	-0.2399	0.41373	13.33879
B_application	spread between the stations		0.20484	0.05354	0.10447	1.95226

25 Static polar of the model in a slow sweep, Case A: tunnel condition, sampled (one row in 4 of 101); the model's quasi-steady limit, not a measurement

Source: 05_solution/model_static_polar_A_validation.csv. 26 rows.

alpha deg	CL	CD	CM c4	CN	CC	f separation	outside static data	mach M
-5	-0.59527	0.01202	-0.01277	-0.59342	0.04708	0.93842	1	0.3
-4	-0.49723	0.01034	-0.01002	-0.49623	0.03155	0.99547	1	0.3
-3	-0.3815	0.00897	-0.00806	-0.38107	0.0182	0.99983	1	0.3
-2	-0.26475	0.00799	-0.00672	-0.26462	0.00845	1	1	0.3
-1	-0.14804	0.0074	-0.00604	-0.14802	0.00238	1	1	0.3
0	-0.03134	0.0072	-0.00576	-0.03134	0	1	0	0.3
1	0.08542	0.00738	-0.00542	0.08541	0.00131	1	0	0.3
2	0.2023	0.00795	-0.00498	0.2022	0.00631	1	0	0.3
3	0.31939	0.0089	-0.00406	0.31905	0.01502	1	0	0.3
4	0.43674	0.01022	-0.00253	0.43589	0.02745	0.99982	0	0.3
5	0.55008	0.0119	0.00026	0.5484	0.04326	0.98387	0	0.3
6	0.65031	0.01382	0.00307	0.64744	0.06139	0.93352	0	0.3

alpha deg	CL	CD	CM c4	CN	CC	f separation	outside static data	mach M
7	0.76719	0.01622	0.00542	0.76257	0.08454	0.9387	0	0.3
8	0.88789	0.01903	0.00767	0.8809	0.11185	0.94948	0	0.3
9	1.00308	0.02214	0.0099	0.99306	0.14216	0.94579	0	0.3
10	1.11301	0.02551	0.01241	1.09928	0.17524	0.9325	0	0.3
11	1.19754	0.02888	0.01661	1.17967	0.20722	0.88149	0	0.3
12	1.29215	0.03494	0.02116	1.26968	0.24152	0.85143	0	0.3
13	1.36174	0.05592	0.02762	1.3378	0.25885	0.79297	0	0.3
14	1.40035	0.09273	0.02409	1.37944	0.25578	0.71344	0	0.3
15	1.07251	0.18584	-0.06812	1.0822	0.10503	0.33788	0	0.3
16	0.90967	0.22976	-0.08939	0.93578	0.0368	0.18004	0	0.3
17	0.8465	0.25332	-0.08394	0.88147	0.01213	0.11106	0	0.3
18	0.82404	0.27002	-0.07937	0.86493	0.00468	0.08368	0	0.3
19	0.82888	0.29058	-0.08214	0.87599	0.00192	0.06913	0	0.3
20	0.85086	0.31593	-0.09115	0.90514	0.0009	0.06306	0	0.3

26 Static polar of the model in a slow sweep, Case B: rotor-blade-station condition, sampled (one row in 4 of 101); the model's quasi-steady limit, not a measurement

Source: 05_solution/model_static_polar_B_application.csv. 26 rows.

alpha deg	CL	CD	CM c4	CN	CC	f separation	outside static data	mach M
-5	-0.59765	0.01204	-0.01155	-0.5958	0.04727	0.9636	1	0.28
-4	-0.49134	0.0103	-0.00917	-0.49036	0.03118	0.99513	1	0.28
-3	-0.37525	0.00894	-0.00755	-0.37482	0.0179	0.9988	1	0.28
-2	-0.25846	0.00797	-0.00631	-0.25833	0.00825	0.99999	1	0.28
-1	-0.14151	0.00739	-0.00561	-0.14149	0.00228	1	1	0.28
0	-0.02457	0.0072	-0.00521	-0.02457	0	1	0	0.28
1	0.09242	0.00739	-0.00465	0.09241	0.00142	1	0	0.28
2	0.20954	0.00797	-0.00413	0.20944	0.00654	1	0	0.28
3	0.32687	0.00893	-0.00315	0.32651	0.01538	1	0	0.28

alpha deg	CL	CD	CM c4	CN	CC	f separation	outside static data	mach M
4	0.44416	0.01027	-0.00163	0.44329	0.02792	0.99847	0	0.28
5	0.5609	0.01198	0.00059	0.55918	0.04412	0.99429	0	0.28
6	0.66345	0.01394	0.00349	0.66052	0.06265	0.94866	0	0.28
7	0.77625	0.01632	0.00655	0.77158	0.08555	0.94087	0	0.28
8	0.89477	0.01912	0.00847	0.88772	0.11272	0.94605	0	0.28
9	1.01002	0.02224	0.01062	0.99994	0.14315	0.94246	0	0.28
10	1.12229	0.02565	0.01307	1.10845	0.17671	0.93318	0	0.28
11	1.21876	0.02919	0.01637	1.20056	0.21097	0.90018	0	0.28
12	1.30323	0.03289	0.02053	1.28009	0.24583	0.8554	0	0.28
13	1.39423	0.04806	0.02491	1.36769	0.27383	0.82258	0	0.28
14	1.44288	0.07688	0.0276	1.41688	0.28145	0.75473	0	0.28
15	1.2636	0.16894	-0.0418	1.26241	0.17081	0.46592	0	0.28
16	0.95688	0.2275	-0.07327	0.98053	0.05198	0.204	0	0.28
17	0.87049	0.25426	-0.08758	0.90469	0.01824	0.12448	0	0.28
18	0.8239	0.26885	-0.08005	0.86443	0.00576	0.08178	0	0.28
19	0.82731	0.2897	-0.08089	0.87421	0.00224	0.06554	0	0.28
20	0.83561	0.31038	-0.08276	0.88891	0.0009	0.05606	0	0.28

27 Convergence of the calibration loops

Source: results/convergence_summary.csv. 2 rows.

study	default compared with	n frames	worst change in loop error	worst change in CL	worst change in CM	worst change in CD	worst error against zero step	target change in loop error	met
step size	4 times as many steps	20	0.000758	0.006153	0.002298	0.001718	0.000602	0.002	yes
cycles marched	4 more cycles	20	0	0	0	0		0.002	yes

28 Response surface (a section in a steady stream with prescribed pitch; damping is model output only)

Source: 05_solution/response_surface.csv. 42 rows.

alpha mean deg	reduced freq k	alpha amp deg	mach M	peak alpha deg	steps per cycle	cycles	CL max	CM min	cycle damping Xi	stall onset alpha deg
6	0.04	8	0.28	14	7920	3	1.48533	-0.01014	0.0702	12.87
6	0.06	8	0.28	14	5400	3	1.49885	-0.0134	0.10118	13.216
6	0.08	8	0.28	14	3960	4	1.49489	-0.01668	0.12824	13.513
6	0.1	8	0.28	14	3240	5	1.48104	-0.01993	0.15411	13.747
6	0.12	8	0.28	14	2880	5	1.46219	-0.02311	0.18029	13.913
6	0.14	8	0.28	14	2520	6	1.44133	-0.02621	0.20502	13.996
6	0.16	8	0.28	14	2160	7	1.423	-0.02921	0.22932	13.943
8	0.04	8	0.28	16	7920	3	1.57706	-0.15696	0.0465	13.067
8	0.06	8	0.28	16	5400	3	1.62695	-0.07831	0.06737	13.5
8	0.08	8	0.28	16	3960	4	1.6512	-0.09925	0.09341	13.883
8	0.1	8	0.28	16	3240	5	1.66219	-0.09942	0.12258	14.213
8	0.12	8	0.28	16	2880	5	1.66312	-0.05241	0.15346	14.496
8	0.14	8	0.28	16	2520	6	1.65609	-0.03286	0.17797	14.737
8	0.16	8	0.28	16	2160	7	1.64641	-0.02613	0.21886	14.943
10	0.04	8	0.28	18	7920	3	1.62201	-0.15584	0.02814	13.147
10	0.06	8	0.28	18	5400	3	1.68307	-0.16771	0.04143	13.597
10	0.08	8	0.28	18	3960	4	1.72356	-0.21517	0.06378	13.987
10	0.1	8	0.28	18	3240	5	1.7496	-0.23942	0.06618	14.318
10	0.12	8	0.28	18	2880	5	1.76187	-0.24181	0.05907	14.594
10	0.14	8	0.28	18	2520	6	1.76633	-0.21303	0.05095	14.827
10	0.16	8	0.28	18	2160	7	1.76578	-0.19193	0.04231	15.022
12	0.04	8	0.28	20	7920	3	1.6432	-0.15239	0.02585	13.15
12	0.06	8	0.28	20	5400	3	1.70959	-0.19592	0.04053	13.577
12	0.08	8	0.28	20	3960	4	1.75607	-0.24927	0.0719	13.934
12	0.1	8	0.28	20	3240	5	1.79164	-0.28469	0.08722	14.222
12	0.12	8	0.28	20	2880	5	1.80486	-0.29176	0.07742	14.45
12	0.14	8	0.28	20	2520	6	1.80749	-0.29068	0.05449	14.628

alpha mean deg	reduced freq k	alpha amp deg	mach M	peak alpha deg	steps per cycle	cycles	CL max	CM min	cycle damping Xi	stall onset alpha deg
12	0.16	8	0.28	20	2160	7	1.80422	-0.27955	0.02528	14.766
14	0.04	8	0.28	22	7920	3	1.64903	-0.16046	0.02274	13.085
14	0.06	8	0.28	22	5400	3	1.71749	-0.2025	0.03622	13.456
14	0.08	8	0.28	22	3960	4	1.76664	-0.26604	0.07162	13.744
14	0.1	8	0.28	22	3240	5	1.79728	-0.29368	0.10473	13.955
14	0.12	8	0.28	22	2880	5	1.83343	-0.30901	0.12219	14.1
14	0.14	8	0.28	22	2520	6	1.8325	-0.31428	0.1122	14.191
14	0.16	8	0.28	22	2160	7	1.8248	-0.31454	0.08933	14.238
16	0.04	8	0.28	24	7920	3	1.63802	-0.17339	0.02826	12.945
16	0.06	8	0.28	24	5400	3	1.71067	-0.18994	0.03875	13.222
16	0.08	8	0.28	24	3960	4	1.75478	-0.25385	0.0634	13.403
16	0.1	8	0.28	24	3240	5	1.77947	-0.28852	0.10726	13.497
16	0.12	8	0.28	24	2880	5	1.79973	-0.30373	0.14296	13.517
16	0.14	8	0.28	24	2520	6	1.84245	-0.32235	0.17081	13.479
16	0.16	8	0.28	24	2160	7	1.83166	-0.32768	0.16979	13.393

29 The reconstruction instant by instant, Case A: tunnel condition (not a flow solution)

Source: 05_solution/reconstruction_A_validation.csv. 6 rows.

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[illegible]

Source: 05_solution/reconstruction_B_application.csv. 6 rows.

[illegible]

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31 State-space form against the indicial march

Source: results/statespace_summary.csv. 17 rows.

quantity	value
n_frames	24
n_frames_below_mach_0_20	4
lowest_mach	0.035
states	14
integrator	classical fourth-order Runge-Kutta, fixed step
worst_difference_in_loop_error	0.000398
mean_difference_in_loop_error	7.2×10^{-5}
worst_difference_in_CL	0.005225
worst_difference_in_CM	0.004843
worst_difference_in_CD	0.001692
worst_difference_in_CL_pct_of_range	0.5275
worst_difference_in_CM_pct_of_range	1.9679
worst_difference_in_CD_pct_of_range	3.6068
worst_difference_in_damping	0.00027
cpu_ratio_state_space_to_indicial	4.94
target_difference_in_loop_error	0.002
met	yes

32 A section free in torsion: the two forms of the model (model output only)

Source: results/structural_checks.csv. 6 rows.

alpha mean deg	alpha amp deg	k	M	cycle damping	cycle work J per m	work from damping J per m	work identity error pct	free log growth per cycle	predicted log growth per cycle	free over predicted growth	free amplitude first deg	free amplitude last deg	free mean drift deg	free log growth per cycle state space	state space agrees in sign	free response	prescribed damping	agree	dimensional identity max abs	peak lift N per m	min moment Nm per m
5	5	0.1	0.3	0.1575	-7.3044	-7.3043	0.0013	-0.00833	-0.00824	1.011	4.99	3.92	0.008	-0.00833	yes	decays	positive	yes	$5.551115123 \times 10^{-16}$	3384.1	-27.89
9	5	0.1	0.3	0.1741	-8.0762	-8.0759	0.0044	-0.00896	-0.00912	0.983	4.987	3.842	0.009	-0.00897	yes	decays	positive	yes	$6.661338148 \times 10^{-16}$	4703.9	-11.84

alpha mean deg	alpha amp deg	k	M	cycle da mping	cycle work J per m	work from damping J per m	work identity error pct	free log growth per cycle	predicted log growth per cycle	free over predicted growth	free amplitude first deg	free amplitude last deg	free mean drift deg	free log growth per cycle state space	state space agrees in sign	free resp onse	prescribed damping	agre e	dimensional identity max abs	peak lift N per m	min moment Nm per m
11	5	0.1	0.3	0.048	-2.2277	-2.2277	0.0029	-0.00268	-0.00251	1.066	4.99	4.619	0.01	-0.00274	yes	decays	positive	yes	8.881784197 × 10 ⁻¹⁶	5118.1	-259.46
12	5	0.05	0.3	0.0284	-1.318	-1.3231	0.3814	-0.00139	-0.00149	0.934	4.987	4.785	0.01	-0.00146	yes	decays	positive	yes	8.881784197 × 10 ⁻¹⁶	4867	-382.56
15	5	0.1	0.29	-0.0506	2.195	2.1954	0.0153	0.00159	0.00265	0.601	5.009	5.267	-0.001	0.00155	yes	grows	negative	yes	9.992007222 × 10 ⁻¹⁶	4887.7	-278.39
10	8	0.15	0.25	0.0532	-4.3837	-4.3763	0.17	-0.00384	-0.00278	1.381	7.976	7.148	0.01	-0.00394	yes	decays	positive	yes	1.1379786 × 10 ⁻¹⁵	4021.4	-258.82

33 Calibration cost against the stall-onset level

Source: results/onset_level_check.csv. 5 rows.

onset level factor	cost	mean nRMS CL	mean nRMS CM	mean nRMS CD
0.92	2.07063	0.0806	0.195	0.1901
0.96	1.99664	0.0804	0.1907	0.1874
1	1.90576	0.082	0.1866	0.1835
1.04	2.02441	0.0845	0.1957	0.1871
1.08	2.08176	0.0847	0.2021	0.1891

34 Runtime environment

Source: 05_solution/runtime_environment.csv. 14 rows.

property	value
python	3.12.3
platform	Linux-6.18.33.1-microsoft-standard-WSL2-x86_64-with-glibc2.39
processor	x86_64
version_numpy	1.26.4
version_scipy	1.11.4
version_pandas	3.0.3

property	value
version_matplotlib	3.6.3
steps_per_cycle_A_validation	3240
cycles_A_validation	5
march_cpu_time_s_A_validation	0.489
steps_per_cycle_B_application	4320
cycles_B_application	4
march_cpu_time_s_B_application	0.453
timing_note	CPU time of the load-model march on this machine; it varies from run to run

35 Accuracy on the held-out loops

Source: results/tables.md, Table 5. Accuracy on the held-out loops. 6 rows.

Measure	Loops	Measured	95 % interval	Target	Met
Loop error, lift	45	0.107	0.091 to 0.124	at most 0.10	no
Loop error, moment	45	0.196	0.173 to 0.220	at most 0.15	no
Loop error, drag	45	0.153	0.131 to 0.178	at most 0.20	yes
Incidence of maximum lift, deg	45	0.572	0.413 to 0.774	at most 0.72	yes
Incidence of moment stall, deg	29	0.622	0.392 to 0.884	at most 1.00	yes
Fraction of loops with the damping sign correct	45	0.600	0.466 to 0.733	at least 0.80	no

36 Held-out measures against their targets

Source: results/validation_targets.csv. 6 rows.

measure	target	kind	measured	met	note
mean_nRMS_CL	0.1	at most	0.1067	no	
mean_nRMS_CM	0.15	at most	0.1959	no	
mean_nRMS_CD	0.2	at most	0.1526	yes	
mean_abs_dalpha_CLmax_deg	0.7238	at most	0.5723	yes	

measure	target	kind	measured	met	note
mean_abs_dalpha_Mstall_deg	1	at most	0.6216	yes	judged on the 29 loops where measurement and model both show a moment stall; the model misses a measured one on 3 loops and predicts an unmeasured one on 0
damping_sign_agreement	0.8	at least	0.6	no	

37 Held-out accuracy by stall depth

Source: results/validation_by_stall_depth.csv. 3 rows.

group	n frames	nRMS CL	nRMS CM	nRMS CD	damping sign agreement	mean abs measured damping
attached or marginal (peak less than 2 deg beyond static stall)	19	0.0927	0.2407	0.209	0.4737	0.0983
light stall (2 to 6 deg beyond)	7	0.1306	0.1553	0.1517	0.4286	0.0563
deep stall (6 deg or more beyond)	19	0.112	0.166	0.0964	0.7895	0.1246

38 Cycle damping on the held-out loops (the model does not predict cycle damping through stall)

Source: results/validation_damping.csv. 6 rows.

group	n frames	measured negative	predicted negative	sign agreement	sign agreement if always positive	correlation	rms error	mean error	mean measured	mean predicted	mean uncertainty random	mean uncertainty stroke
all primary held-out loops with a closed moment loop	45	15	3	0.6	0.6667	0.33	0.1202	0.029	0.064	0.093	0.0152	0.0845
measured damping positive	30	0	3	0.9	1	0.522	0.0893	-0.0302	0.1251	0.0949	0.0135	0.0777
measured damping negative	15	15	0	0	0	0.241	0.1654	0.1475	-0.0583	0.0892	0.0185	0.0982
deep stall (6 deg or more beyond static stall)	19	3	1	0.7895	0.8421	0.554	0.0902	-0.0477	0.1147	0.067	0.0104	0.0659
measured sign certain if the moment error is random (beyond 2 sigma)	35	9	3	0.6571	0.7429	0.435	0.1144	0.0066	0.0825	0.0891	0.0144	0.0811

group	n frames	measured negative	predicted negative	sign agreement	sign agreement if always positive	correlation	rms error	mean error	mean measured	mean predicted	mean uncertainty random	mean uncertainty stroke
measured sign certain even if the moment error follows the stroke	21	3	0	0.8571	0.8571	0.684	0.0986	-0.026	0.1317	0.1057	0.0116	0.0693

39 Tabulated against fitted separation law, same constants

Source: results/comparison_same_constants.csv. 21 rows.

constants	metric	tabulated	reference	paired difference	ci95 lo	ci95 hi	n frames	verdict
literature values	mean nRMS_CL	0.1332	0.1315	0.0017	0	0.0033	45	reference better
literature values	mean nRMS_CM	0.2161	0.2145	0.0017	0.0002	0.0034	45	reference better
literature values	mean nRMS_CD	0.1641	0.1633	0.0008	-0.0004	0.0021	45	no difference at 95 %
literature values	mean CLmax_err_pct	4.7574	4.8025	-0.0451	-0.1834	0.1085	45	no difference at 95 %
literature values	mean dalpha_CLmax_deg	0.8938	0.9116	-0.0178	-0.0697	0.0183	45	no difference at 95 %
literature values	mean dalpha_Mstall_deg	0.7817	0.7871	-0.0054	-0.0107	-0.0009	30	tabulated better
literature values	damping sign agreement	0.6444	0.6444	0			45	
fitted for the tabulated model	mean nRMS_CL	0.1067	0.1054	0.0014	-0.0006	0.003	45	no difference at 95 %
fitted for the tabulated model	mean nRMS_CM	0.1959	0.1945	0.0014	0.0002	0.0027	45	reference better
fitted for the tabulated model	mean nRMS_CD	0.1526	0.152	0.0005	-0.0007	0.0016	45	no difference at 95 %
fitted for the tabulated model	mean CLmax_err_pct	4.5311	4.4242	0.1069	0.0051	0.2133	45	reference better
fitted for the tabulated model	mean dalpha_CLmax_deg	0.5723	0.5628	0.0095	-0.0298	0.0433	45	no difference at 95 %
fitted for the tabulated model	mean dalpha_Mstall_deg	0.6216	0.6228	-0.0011	-0.0028	0.0004	29	no difference at 95 %
fitted for the tabulated model	damping sign agreement	0.6	0.6	0			45	
fitted for the reference model	mean nRMS_CL	0.1058	0.1056	0.0002	-0.0032	0.0027	45	no difference at 95 %
fitted for the reference model	mean nRMS_CM	0.1969	0.1963	0.0006	-0.0015	0.0024	45	no difference at 95 %
fitted for the reference model	mean nRMS_CD	0.151	0.1512	-0.0002	-0.0028	0.0016	45	no difference at 95 %
fitted for the reference model	mean CLmax_err_pct	4.4817	4.3475	0.1342	0.0278	0.243	45	reference better
fitted for the reference model	mean dalpha_CLmax_deg	0.5295	0.5392	-0.0097	-0.0427	0.0103	45	no difference at 95 %
fitted for the reference model	mean dalpha_Mstall_deg	0.6442	0.6476	-0.0034	-0.0078	-0.0003	29	tabulated better

constants	metric	tabulated	reference	paired difference	ci95 lo	ci95 hi	n frames	verdict
fitted for the reference model	damping sign agreement	0.6222	0.6222	0			45	

40 Every target against its measured value

Source: results/targets_scoreboard.csv. 34 rows.

group	kind of target	target	kind	value	measured	measurement floor	met	source
attached_flow	physical	lift_and_moment_amplitude_error_max_pct	at most	3	1.86		yes	attached_checks.csv
attached_flow	physical	lift_and_moment_phase_error_max_deg	at most	3	0.62		yes	attached_checks.csv
attached_flow	physical	chord_speed_independence_max_abs_dCL	at most	1×10^{-10}	5.04×10^{-14}		yes	attached_checks.csv
attached_flow	physical	no_stall_frame_nRMS_CL_max	at most	0.05	0.0425		yes	attached_checks.csv
attached_flow	physical	no_stall_frame_nRMS_CM_max	at most	0.1	0.1235		no	attached_checks.csv
separated_flow	physical	CD_negative_fraction_of_cycle_max	at most	0.05	0		yes	separated_checks.csv
separated_flow	physical	CD_min	at least	-0.01	0.0021		yes	separated_checks.csv
separated_flow	physical	onset_shift_with_4x_steps_max_deg	at most	0.02	0.0011		yes	separated_checks.csv
calibration	numerical	repeatability_max_abs_difference	at most	1×10^{-6}	1.11×10^{-16}		yes	calibration_repeatability.json
held_out_validation	physical	mean_nRMS_CL_max	at most	0.1	0.1067	0.0279	no	validation_targets.csv
held_out_validation	physical	mean_nRMS_CM_max	at most	0.15	0.1959	0.0589	no	validation_targets.csv
held_out_validation	physical	mean_nRMS_CD_max	at most	0.2	0.1526	0.059	yes	validation_targets.csv
held_out_validation	physical	mean_abs_dalpha_CLmax_deg_max	at most	0.7238	0.5723		yes	validation_targets.csv
held_out_validation	physical	mean_abs_dalpha_Mstall_deg_max	at most	1	0.6216		yes	validation_targets.csv: judged on the 29 loops where measurement and model both show a moment stall; the model misses a measured one on 3 loops and predicts an unmeasured one on 0

group	kind of target	target	kind	value	measured	measurement floor	met	source
held_out_validation	physical	damping_sign_agreement_min	at least	0.8	0.6		no	validation_targets.csv
structural	numerical	cycle_work_identity_max_error_pct	at most	1	0.3814		yes	structural_checks.csv
structural	numerical	free_response_sign_agreement_min_conditions	at least	3	6		yes	structural_checks.csv
structural	numerical	dimensional_load_identity_max_abs	at most	1×10^{-9}	1.138×10^{-15}		yes	structural_checks.csv
structural	numerical	second_axis_theodorsen_amplitude_max_pct	at most	3	1.69		yes	attached_checks.csv
structural	numerical	second_axis_theodorsen_phase_max_deg	at most	3	0.63		yes	attached_checks.csv
code_quality	software	regression_max_abs_difference	at most	1×10^{-12}	0		yes	code_quality.csv
code_quality	software	import_path_edits_load_model_path_max	at most	0	0		yes	code_quality.csv
code_quality	software	module_level_executable_lines_per_load_model_file_max	at most	30	1		yes	code_quality.csv
code_quality	software	longest_package_function_lines_max	at most	80	56		yes	code_quality.csv
code_quality	software	unused_parameters_max	at most	0	1		no	code_quality.csv
code_quality	software	public_functions_typed_and_unit_documented_pct_min	at least	100	100		yes	code_quality.csv
code_quality	software	lint_findings_max	at most	0	0		yes	code_quality.csv
code_quality	software	silently_skipped_checks_max	at most	0	0		yes	code_quality.csv
code_quality	software	revision_history_phrases_in_package_max	at most	0	0		yes	code_quality.csv
numerical	numerical	loop_error_change_max	at most	0.002	0.000602		yes	convergence_summary.csv
numerical	numerical	state_space_loop_error_difference_max	at most	0.002	0.000398		yes	statespace_summary.csv
equivalence_of_separation_laws	physical	margin_CL	at most	0.01	0.01402		no	equivalence.csv
equivalence_of_separation_laws	physical	margin_CM	at most	0.015	0.01143		yes	equivalence.csv
equivalence_of_separation_laws	physical	margin_CD	at most	0.02	0.01159		yes	equivalence.csv

41 Scored loops, one row each (the same numbers as results/validation_table.csv)

Source: 06_postprocessing/validation/validation_loops_summary.csv. 84 rows.

frame	set	M	k	alpha0 deg	amp deg	nRMS CL	nRMS CM	nRMS CD	CLmax err pct	dalpha CLmax deg	dalpha Mstall deg	Xi exp	Xi model	Xi sign agree	in mach range
frame_7019	held_out	0.297	0.051	9	4.9	0.048	0.2364	0.1635	0.7961	0.1901		0.0465	0.1016	1	yes
frame_7021	held_out	0.299	0.1	9	4.9	0.1059	0.364	0.3047	-0.4731	0.2233		-0.1012	0.1733	0	yes
frame_7023	calibration	0.299	0.201	9	5	0.1381	0.3592	0.3381	-8.9657	0.021		0.1121	0.2836	1	yes
frame_7101	calibration	0.301	0.15	9	4.9	0.0972	0.3827	0.297	-5.0528	0.1402		0.0402	0.2261	1	yes
frame_7104	calibration	0.301	0.025	8.9	4.9	0.042	0.0751	0.2589	0.6245	0.5236		0.0341	0.0535	1	yes
frame_7108	held_out	0.301	0.025	8	4.9	0.0294	0.1557	0.4392	-0.6438	0.2097		0.0341	0.0497	1	yes
frame_7110	calibration	0.301	0.1	8	4.9	0.0425	0.1249	0.2056	-2.5241	0.1281		0.16	0.1676	1	yes
frame_7111	held_out	0.301	0.199	8	5	0.0767	0.1313	0.1556	-8.0406	0.0449		0.3521	0.2866	1	yes
frame_7112	held_out	0.301	0.025	9.9	4.9	0.0759	0.1661	0.1043	1.2295	0.9623	-0.0138	-0.0577	0.0312	0	yes
frame_7113	held_out	0.301	0.099	10	4.9	0.1615	0.3082	0.2083	1.1407	0.5409		-0.0206	0.155	0	yes
frame_7114	held_out	0.301	0.199	10	5	0.2272	0.3409	0.3121	-6.7492	0.5629		0.0354	0.2705	1	yes
frame_7117	held_out	0.301	0.025	10.9	4.9	0.0918	0.2194	0.1272	2.918	0.3863	-0.1321	-0.1106	0.0217	0	yes
frame_7118	held_out	0.301	0.05	10.9	4.9	0.0653	0.1415	0.075	-0.3496	-0.8569	-0.0059	-0.037	0.0374	0	yes
frame_7119	calibration	0.301	0.099	10.9	4.9	0.0879	0.1125	0.1807	1.3966	0.8692	0.0265	0.002	0.0532	1	yes
frame_7120	held_out	0.301	0.149	10.9	4.9	0.1727	0.1108	0.1957	-0.6656	0.8064	-0.0294	0.026	0.0728	1	yes
frame_7121	held_out	0.301	0.198	11	5	0.2096	0.2084	0.2541	-5.4158	0.0068		0.0569	0.0737	1	yes
frame_7200	calibration	0.301	0.025	11.9	4.9	0.1386	0.1491	0.1144	3.886	-0.565	-0.101	-0.0418	0.0171	0	yes
frame_7202	held_out	0.302	0.05	11.93	4.89	0.1284	0.1625	0.0817	1.8917	-0.8923	-0.1186	0.0144	0.027	1	yes
frame_7205	held_out	0.302	0.099	11.9	4.9	0.0915	0.1203	0.1445	1.7059	-0.2953	-0.1159	0.0496	-0.0026	0	yes
frame_7207	held_out	0.302	0.198	11.9	4.9	0.1547	0.1241	0.184	-10.2849	-0.2696	-0.0627	0.0994	-0.0529	0	yes
frame_7212	calibration	0.302	0.149	8.7	5	0.0657	0.2273	0.1881	-3.5555	0.081		0.1542	0.2266	1	yes
frame_7214	held_out	0.302	0.099	8.7	5	0.0421	0.2045	0.1496	-0.3263	0.0473		0.1154	0.1711	1	yes
frame_7216	held_out	0.302	0.05	8.7	5	0.0398	0.0901	0.1907	1.3198	0.1063		0.0815	0.0999	1	yes
frame_7222	held_out	0.298	0.051	10	5.1	0.0958	0.3011	0.1577	2.6638	0.396	-0.1281	-0.1369	0.0539	0	yes
frame_7300	held_out	0.3	0.151	10	5	0.165	0.3372	0.2649	-2.1229	0.3787		-0.0127	0.2223	0	yes
frame_7305	calibration	0.3	0.151	12	4.9	0.085	0.0957	0.1321	-0.7751	1.0781	-0.1634	0.0359	-0.0107	0	yes
frame_8019	calibration	0.035	0.104	10	9.9	0.1434	0.2812	0.2658	-5.4499	-0.2445	1.211	0.0641	0.1403	1	no
frame_8021	held_out	0.035	0.151	10	10	0.1041	0.3005	0.242	-7.8181	-0.7484	0.4581	-0.0298	0.1379	0	no

frame	set	M	k	alpha0 deg	amp deg	nRMS CL	nRMS CM	nRMS CD	CLmax err pct	dalpha CLmax deg	dalpha Mstall deg	Xi exp	Xi model	Xi sign agree	in mach range
frame_8023	held_out	0.035	0.253	10	9.9	0.0834	0.4132	0.3328	-6.5928	-0.2442		-0.1425	0.3124	0	no
frame_8102	held_out	0.036	0.103	15	9.9	0.1897	0.1426	0.1533	-4.8242	-1.2439	0.3659	0.2189	0.2408	1	no
frame_8104	held_out	0.036	0.153	15	9.9	0.1676	0.1436	0.1391	-5.3639	-1.5116	-0.5118	0.074	0.266	1	no
frame_8114	held_out	0.072	0.099	15	9.9	0.1784	0.2266	0.2152	-9.3068	-4.1386	-1.6211	0.0738	0.2315	1	no
frame_8116	held_out	0.072	0.149	15	10	0.1478	0.222	0.2081	-14.5682	-3.571	-2.9495	-0.1099	0.2684	0	no
frame_8118	calibration	0.072	0.248	15	9.9	0.1769	0.2869	0.3033	-15.4569	-2.5242		-0.3091	0.1744	0	no
frame_8203	held_out	0.072	0.248	9.9	9.9	0.1009	0.5446	0.5324	8.4194	-0.1242		0.143	0.3118	1	no
frame_8210	held_out	0.109	0.25	9.9	9.9	0.0781	0.3414	0.5754	4.2458	-0.039		0.3049	0.3021	1	no
frame_8214	calibration	0.109	0.1	15	9.9	0.2111	0.2783	0.2448	-12.9722	-4.6979	-3.3077	-0.0192	0.2246	0	no
frame_8220	calibration	0.184	0.099	14.9	9.9	0.2158	0.2883	0.2436	-11.7271	-4.7542	-4.1331	-0.0592	0.1584	0	no
frame_8222	held_out	0.184	0.149	15	9.9	0.1627	0.2829	0.2259	-14.8491	-3.4153	-4.0731	-0.1792	0.1351	0	no
frame_9022	held_out	0.184	0.235	15	6	0.237	0.3778	0.4381	-11.8705	-1.1741		-0.5503	-0.2143	1	no
frame_9101	held_out	0.184	0.284	15	4.9	0.3013	2.4917	2.0246	-8.7271	-1.5707		0.0002	-0.3423	0	no
frame_9106	held_out	0.184	0.245	9.8	9.9	0.0649	0.3659	0.4433	-2.9088	-0.0227		0.3	0.2044	1	no
frame_9110	held_out	0.184	0.01	8	10	0.1951	0.2771	0.1843	-2.2447	-0.7603	-1.0071	-0.0688	0.0148	0	no
frame_9112	held_out	0.184	0.05	7.9	10	0.1476	0.3105	0.2714	4.3985	-0.6059		-0.0169	0.0516	0	no
frame_9118	held_out	0.184	0.199	7.9	10	0.0358	0.12	0.1773	3.2288	0.1152		0.3076	0.2633	1	no
frame_9202	held_out	0.22	0.098	15	9.9	0.1578	0.2721	0.1914	-10.128	-3.7518	-2.671	0.0383	0.1461	1	yes
frame_9203	held_out	0.249	0.098	15	10	0.088	0.1809	0.1045	-9.3949	-2.03	-1.8194	0.1841	0.1362	1	yes
frame_9208	held_out	0.28	0.097	15	9.9	0.074	0.1229	0.0799	-6.6622	-0.7014	-0.9573	0.2764	0.1274	1	yes
frame_9210	calibration	0.295	0.01	15	10	0.0848	0.1892	0.0785	2.8192	0.5564	-0.1543	-0.0101	0.0142	0	yes
frame_9213	calibration	0.294	0.024	15	10	0.083	0.1009	0.079	2.5327	-0.283	-0.2067	0.054	0.038	1	yes
frame_9214	held_out	0.292	0.049	15	10	0.0894	0.1669	0.1073	0.066	-0.7097	-0.4689	0.1436	0.0467	1	yes
frame_9217	held_out	0.29	0.098	15	10	0.0738	0.134	0.0661	-10.9221	-0.0431	-0.9712	0.2684	0.1286	1	yes
frame_9218	held_out	0.283	0.151	14.9	9.9	0.0889	0.1222	0.0722	-11.5502	-0.7931	-1.8556	0.233	0.1399	1	yes
frame_9221	held_out	0.302	0.01	9.9	9.9	0.0477	0.1893	0.1466	3.4274	0.7467	0.1501	-0.0031	0.0134	0	yes
frame_9222	calibration	0.302	0.024	9.9	9.9	0.0456	0.1382	0.1257	-2.3581	-0.9859	0.1519	0.0179	0.0347	1	yes
frame_9223	held_out	0.302	0.048	10	9.9	0.0457	0.1289	0.1152	-1.5683	0.0001	0.006	0.0712	0.0387	1	yes
frame_9302	calibration	0.302	0.096	9.8	9.9	0.0656	0.1141	0.1135	-3.5891	-0.3253	-0.1739	0.1388	0.0871	1	yes
frame_9307	held_out	0.302	0.145	9.9	9.9	0.1064	0.0799	0.1009	-2.3024	-0.4308	-0.496	0.1433	0.0566	1	yes
frame_10022	held_out	0.301	0.098	12	9.9	0.0863	0.0973	0.0806	-3.8288	-0.4989	-0.5207	0.167	0.1057	1	yes
frame_10104	calibration	0.302	0.048	11.9	7.9	0.0959	0.1432	0.0745	-1.4762	-0.5637	-0.4053	0.0193	0.0228	1	yes
frame_10105	held_out	0.302	0.097	11.9	7.9	0.0678	0.1033	0.084	-0.6	-0.5773	-0.4429	0.1164	0.0827	1	yes

frame	set	M	k	alpha0 deg	amp deg	nRMS CL	nRMS CM	nRMS CD	CLmax err pct	dalpha CLmax deg	dalpha Mstall deg	Xi exp	Xi model	Xi sign agree	in mach range
frame_10108	held_out	0.296	0.125	11.9	7.9	0.0713	0.0929	0.076	-9.6744	-1.0603	-0.9048	0.1055	0.0747	1	yes
frame_10113	calibration	0.302	0.01	15	5	0.1159	0.197	0.1506	3.0762	0.5639	-0.3735		0.0096		yes
frame_10114	held_out	0.295	0.025	15	5	0.1276	0.2448	0.1143	0.1785	-0.551	-0.4345	-0.081	0.0279	0	yes
frame_10117	held_out	0.295	0.05	15	5	0.2776	0.3032	0.1061	-4.7433	-0.6918	-0.5942	-0.0101	0.0608	0	yes
frame_10118	held_out	0.291	0.102	15	5	0.2522	0.2215	0.0766	-6.8911	-0.8282	-0.8664	0.0727	-0.0498	0	yes
frame_10120	held_out	0.294	0.151	15	4.9	0.1588	0.2387	0.0898	-11.7015	-0.9146	-1.3247	0.1729	0.0297	1	yes
frame_10123	held_out	0.293	0.202	15	4.9	0.1235	0.1625	0.1007	-19.542	-0.8166	-1.7076	0.0726	0.0203	1	yes
frame_10202	held_out	0.301	0.01	10	4.9	0.0961	0.3149	0.2798	4.1604	0.4591	-0.3846	-0.0587	0.0136	0	yes
frame_10203	calibration	0.301	0.025	10	5	0.0725	0.4577	0.2802	4.4578	0.359		-0.0357	0.0284	0	yes
frame_10204	held_out	0.3	0.049	10	5	0.0794	0.2591	0.1228	1.6662	0.0352	0.0373	-0.1117	0.0541	0	yes
frame_10207	held_out	0.302	0.074	9.9	4.9	0.1049	0.2258	0.1186	2.1866	-0.2137		-0.0492	0.0919	0	yes
frame_10208	held_out	0.3	0.099	10	4.9	0.116	0.3	0.1794	1.4683	0.1988		-0.0347	0.1584	0	yes
frame_10211	held_out	0.3	0.149	9.97	4.9	0.1525	0.3402	0.2588	-1.6087	0.2435		-0.0498	0.2229	0	yes
frame_10212	calibration	0.3	0.198	10	5	0.209	0.3344	0.3208	-5.9909	0.4816		-0.0202	0.2691	0	yes
frame_10218	calibration	0.3	0.01	5	5	0.0304	0.1002	0.219	-0.7487	0.132			0.0172		yes
frame_10221	calibration	0.301	0.099	5	5	0.0339	0.1202	0.2103	-4.7122	0.0169		0.1395	0.1562	1	yes
frame_10222	held_out	0.301	0.198	5	5	0.0578	0.1142	0.1672	-10.8233	0.0375		0.3286	0.2757	1	yes
frame_10303	held_out	0.301	0.099	5	10	0.064	0.2976	0.2528	-0.8427	0.0479		0.0856	0.1514	1	yes
frame_10305	calibration	0.301	0.099	3.7	10	0.0215	0.183	0.2176	-4.6428	-0.1418		0.1263	0.1518	1	yes
frame_10309	held_out	0.301	0.099	3	10	0.0235	0.0865	0.1402	-5.7016	0.223		0.1544	0.1518	1	yes
frame_14218	held_out	0.292	0.025	15	10	0.0984	0.1112	0.0428	-7.4111	-1.3689	-0.2847	0.0537	0.0367	1	yes
frame_14219	held_out	0.291	0.051	15	10	0.0929	0.1818	0.0772	-6.1145	-0.6065	-0.5233	0.1543	0.0502	1	yes
frame_14220	calibration	0.287	0.103	15	10	0.0849	0.1281	0.0849	-9.6114	-0.4423	-1.0573	0.2855	0.1359	1	yes

42 Constants of the calibration, fitted and fixed

Source: 06_postprocessing/validation/calibration_constants.csv. 16 rows.

constant	value	status	literature value	lower bound	upper bound
Tp	1.651526	fitted	1.7	0.3	4
Tf0	6.278239	fitted	3	0.5	10
Tv0	6.317996	fitted	6	1	15
Tvl	8.391549	fitted	7	3	13
eta	0.906958	fitted	0.95	0.6	1
xcp_v	0.2	fixed	0.2		
St_sh	0.19	fixed	0.19		
cm_lag	0.1	fixed	0.1		
CD0	0.0072	fixed	0.0072		
cn1_factor	1	fixed	1		
cmq_kirchhoff	1	fixed	1		
cm_unsteady	0.728	fixed	1		
sigma_rules	1	fixed	1		
cc_geometric	1	fixed	1		
cc_offset	0	fixed	0		
cc_floor	1	fixed	1		

43 Dimensions of the drawings (illustrative)

Source: 08_engineering_drawings/dimensions.csv. 14 rows.

name	value	unit	source
Case A section	NACA 0012	-	NASA TM-84245
Case A chord of the tunnel model	0.61	m	NASA TM-84245
Case A maximum thickness, percent of chord	12	% of chord	NACA 0012 designation
Case A maximum thickness	0.0732	m	derived: 12 % of the Case A chord
Case A station of maximum thickness, fraction of chord	0.2995	chord	four-digit thickness form (01_geometry)

name	value	unit	source
Case A pitch axis from the leading edge, fraction of chord	0.25	chord	NASA TM-84245
Case B rotor radius	8.18	m	Bousman, NASA/TP-2003-212265 (UH-60A main rotor)
Case B rotor radius, in feet	26.83	ft	Bousman, NASA/TP-2003-212265 (UH-60A main rotor)
Case B blade chord	0.527	m	Bousman, NASA/TP-2003-212265 (UH-60A main rotor)
Case B blade chord, in inches	20.76	in	Bousman, NASA/TP-2003-212265 (UH-60A main rotor)
Case B number of blades	4	-	Bousman, NASA/TP-2003-212265 (UH-60A main rotor)
Case B rotor speed	258	rpm	Bousman, NASA/TP-2003-212265 (UH-60A main rotor)
Case B analysis station, fraction of rotor radius	0.75	R	case definition of this study
Root end of the blade as drawn, fraction of rotor radius	0.13	R	Bousman, NASA/TP-2003-212265 (Table 3: root cutout 42.00 in of 322.00 in)

44 What each drawing sheet states

Source: 08_engineering_drawings/sheet_record.csv. 11 rows.

sheet	item	value
1	scale stated in the title block	NOT TO SCALE; VIEWS NOT ALIGNED IN PROJECTION
1	generic body: mast_over_R	0.15
1	generic body: fin_top_over_R	-0.06
2	scale stated in the title block	NOT TO SCALE (ISOMETRIC SKETCH)
2	generic body: mast_over_R	0.15
2	generic body: fin_top_over_R	-0.06
3	scale stated in the title block	PLANFORM 1 : 27.3 ON A3; SECTION A-A ENLARGED
3	section A-A: cutting-plane arrows point	outboard
3	section A-A: leading edge drawn on the	left
3	section A-A: leading edge belongs on the	left
4	scale stated in the title block	1 : 2.26 ON A3

45 Whether the measured loop was drawn with each case

Source: 06_postprocessing/overlay_record.csv. 2 rows.

case	measured frame	drawn	note	M	k	alpha mean deg	alpha amp deg
A_validation	frame_9302	yes	drawn	0.302	0.096	9.8	9.9
B_application		no	the case has no measured loop				

46 The region beyond the critical pressure printed on each figure of a corrected quantity

Source: 06_postprocessing/masked_region_record.csv. 52 rows.

file	phase	surface fraction	field area over c2	field fraction
cp_distribution_A_validation.png	mean_up	0		
cp_distribution_A_validation.png	rise	0.019556	0	0
cp_distribution_A_validation.png	peak	0.032054	0.00086813	0.00012034
cp_distribution_A_validation.png	dsv	0.030873	0.00086813	0.00012034
cp_distribution_A_validation.png	fall	0	0	0
cp_distribution_A_validation.png	mean_down	0		
temperature_profile_A_validation.png	rise	0.019556	0	0
temperature_profile_A_validation.png	peak	0.032054	0.00086813	0.00012034
temperature_profile_A_validation.png	dsv	0.030873	0.00086813	0.00012034
temperature_profile_A_validation.png	fall	0	0	0
contour_Cp_A_validation_rise_a15.png	rise	0.019556	0	0
contour_Mach_A_validation_rise_a15.png	rise	0.019556	0	0
contour_Tstatic_A_validation_rise_a15.png	rise	0.019556	0	0
contour_Trecovery_A_validation_rise_a15.png	rise	0.019556	0	0
contour_Cp_A_validation_peak_a18.png	peak	0.032054	0.00086813	0.00012034
contour_Mach_A_validation_peak_a18.png	peak	0.032054	0.00086813	0.00012034
contour_Tstatic_A_validation_peak_a18.png	peak	0.032054	0.00086813	0.00012034
contour_Trecovery_A_validation_peak_a18.png	peak	0.032054	0.00086813	0.00012034
contour_Cp_A_validation_dsv_a20.png	dsv	0.030873	0.00086813	0.00012034
contour_Mach_A_validation_dsv_a20.png	dsv	0.030873	0.00086813	0.00012034

file	phase	surface fraction	field area over c2	field fraction
contour_Tstatic_A_validation_dsv_a20.png	dsv	0.030873	0.00086813	0.00012034
contour_Trecovery_A_validation_dsv_a20.png	dsv	0.030873	0.00086813	0.00012034
contour_Cp_A_validation_fall_a15.png	fall	0	0	0
contour_Mach_A_validation_fall_a15.png	fall	0	0	0
contour_Tstatic_A_validation_fall_a15.png	fall	0	0	0
contour_Trecovery_A_validation_fall_a15.png	fall	0	0	0
cp_distribution_B_application.png	mean_up	0.0051749		
cp_distribution_B_application.png	rise	0.020276	0	0
cp_distribution_B_application.png	peak	0.022925	0.00057875	8.0225×10^{-5}
cp_distribution_B_application.png	dsv	0.022925	0.00057875	8.0225×10^{-5}
cp_distribution_B_application.png	fall	0	0	0
cp_distribution_B_application.png	mean_down	0		
temperature_profile_B_application.png	rise	0.020276	0	0
temperature_profile_B_application.png	peak	0.022925	0.00057875	8.0225×10^{-5}
temperature_profile_B_application.png	dsv	0.022925	0.00057875	8.0225×10^{-5}
temperature_profile_B_application.png	fall	0	0	0
contour_Cp_B_application_rise_a16.png	rise	0.020276	0	0
contour_Mach_B_application_rise_a16.png	rise	0.020276	0	0
contour_Tstatic_B_application_rise_a16.png	rise	0.020276	0	0
contour_Trecovery_B_application_rise_a16.png	rise	0.020276	0	0
contour_Cp_B_application_peak_a17.png	peak	0.022925	0.00057875	8.0225×10^{-5}
contour_Mach_B_application_peak_a17.png	peak	0.022925	0.00057875	8.0225×10^{-5}
contour_Tstatic_B_application_peak_a17.png	peak	0.022925	0.00057875	8.0225×10^{-5}
contour_Trecovery_B_application_peak_a17.png	peak	0.022925	0.00057875	8.0225×10^{-5}
contour_Cp_B_application_dsv_a18.png	dsv	0.022925	0.00057875	8.0225×10^{-5}
contour_Mach_B_application_dsv_a18.png	dsv	0.022925	0.00057875	8.0225×10^{-5}
contour_Tstatic_B_application_dsv_a18.png	dsv	0.022925	0.00057875	8.0225×10^{-5}
contour_Trecovery_B_application_dsv_a18.png	dsv	0.022925	0.00057875	8.0225×10^{-5}
contour_Cp_B_application_fall_a16.png	fall	0	0	0

file	phase	surface fraction	field area over c2	field fraction
contour_Mach_B_application_fall_a16.png	fall	0	0	0
contour_Tstatic_B_application_fall_a16.png	fall	0	0	0
contour_Trecovery_B_application_fall_a16.png	fall	0	0	0

47 File inventory of 05_solution

Source: 05_solution (sizes read from the files at build time). 36 rows.

File	Size, kB	Rows	Columns	Holds
README.md	1.7			not described in the dossier
convergence/panel_convergence_A_validation.csv	3.5	24	15	the reconstruction at the six instants against the number of panels
convergence/panel_convergence_B_application.csv	3.5	24	15	the reconstruction at the six instants against the number of panels
convergence/panel_convergence_summary.csv	0.9	8	8	largest change of the quoted quantities of the reconstruction when the panels are doubled, against the stated tolerances
convergence/reconstruction_timing.csv	0.3	2	7	CPU time of the reconstruction of each case
convergence/residuals_A_validation.csv	0.6	8	8	the reported loads and their largest change over the cycle against the number of cycles marched
convergence/residuals_B_application.csv	0.6	8	8	the reported loads and their largest change over the cycle against the number of cycles marched
convergence/timestep_order.csv	1.6	8	13	observed order of convergence, Richardson estimate and discretisation uncertainty of each load
convergence/timestep_refinement.csv	1.6	12	12	loads of each case against the number of steps per cycle
cp_distribution_A_validation.csv	207.5	1920	16	reconstructed surface pressure (corrected and incompressible), local Mach number, static and recovery temperature, speed and transpiration at instants of the cycle, with the flag of the region beyond the critical pressure (reconstruction, not a flow solution)
cp_distribution_B_application.csv	206.3	1920	16	reconstructed surface pressure (corrected and incompressible), local Mach number, static and recovery temperature, speed and transpiration at instants of the cycle, with the flag of the region beyond the critical pressure (reconstruction, not a flow solution)
cycle_scan_A_validation.csv	40.2	361	14	the transpiration, the closure of the lift and the vortex marker at instants spread over the cycle
cycle_scan_B_application.csv	40.1	361	14	the transpiration, the closure of the lift and the vortex marker at instants spread over the cycle

File	Size, kB	Rows	Columns	Holds
field_A_validation_dsv_a20.csv.gz	416.6	25200	12	reconstructed field on the rectangular grid at one instant: incompressible velocity, pressure coefficient (corrected and incompressible), local Mach number, static and recovery temperature, vorticity, the flag of the region beyond the critical pressure (values not physical there) and a flag for the nodes inside the section (reconstruction, not a flow solution), gzip-compressed
field_A_validation_fall_a15.csv.gz	355.1	25200	12	reconstructed field on the rectangular grid at one instant: incompressible velocity, pressure coefficient (corrected and incompressible), local Mach number, static and recovery temperature, vorticity, the flag of the region beyond the critical pressure (values not physical there) and a flag for the nodes inside the section (reconstruction, not a flow solution), gzip-compressed
field_A_validation_peak_a18.csv.gz	415.1	25200	12	reconstructed field on the rectangular grid at one instant: incompressible velocity, pressure coefficient (corrected and incompressible), local Mach number, static and recovery temperature, vorticity, the flag of the region beyond the critical pressure (values not physical there) and a flag for the nodes inside the section (reconstruction, not a flow solution), gzip-compressed
field_A_validation_peak_on_mesh.csv.gz	372.8	15633	9	reconstructed pressure coefficient and speed at the nodes of the O-grid of 02_mesh at the instant of peak lift, with the source of each value (reconstruction, not a flow solution), gzip-compressed
field_A_validation_rise_a15.csv.gz	399.2	25200	12	reconstructed field on the rectangular grid at one instant: incompressible velocity, pressure coefficient (corrected and incompressible), local Mach number, static and recovery temperature, vorticity, the flag of the region beyond the critical pressure (values not physical there) and a flag for the nodes inside the section (reconstruction, not a flow solution), gzip-compressed
field_B_application_dsv_a18.csv.gz	411.2	25200	12	reconstructed field on the rectangular grid at one instant: incompressible velocity, pressure coefficient (corrected and incompressible), local Mach number, static and recovery temperature, vorticity, the flag of the region beyond the critical pressure (values not physical there) and a flag for the nodes inside the section (reconstruction, not a flow solution), gzip-compressed
field_B_application_fall_a16.csv.gz	355.2	25200	12	reconstructed field on the rectangular grid at one instant: incompressible velocity, pressure coefficient (corrected and incompressible), local Mach number, static and recovery temperature, vorticity, the flag of the region beyond the critical pressure (values not physical there) and a flag for the nodes inside the section (reconstruction, not a flow solution), gzip-compressed
field_B_application_peak_a17.csv.gz	407.9	25200	12	reconstructed field on the rectangular grid at one instant: incompressible velocity, pressure coefficient (corrected and incompressible), local Mach number, static and recovery temperature, vorticity, the flag of the region beyond the critical pressure (values not physical there) and a flag for the nodes inside the section (reconstruction, not a flow solution), gzip-compressed

File	Size, kB	Rows	Columns	Holds
field_B_application_peak_on_mesh.csv.gz	371.8	15633	9	reconstructed pressure coefficient and speed at the nodes of the O-grid of 02_mesh at the instant of peak lift, with the source of each value (reconstruction, not a flow solution), gzip-compressed
field_B_application_rise_a16.csv.gz	402.0	25200	12	reconstructed field on the rectangular grid at one instant: incompressible velocity, pressure coefficient (corrected and incompressible), local Mach number, static and recovery temperature, vorticity, the flag of the region beyond the critical pressure (values not physical there) and a flag for the nodes inside the section (reconstruction, not a flow solution), gzip-compressed
held_out_error_bands.csv	0.9	5	8	error of the load model in each headline quantity on the held-out loops
metrics_A_validation.csv	5.7	147	3	scalar results of the case
metrics_B_application.csv	5.7	147	3	scalar results of the case
model_static_polar_A_validation.csv	17.1	101	10	lift, drag, moment and separation point of the model in a slow sweep at the case's Mach number: its quasi-steady limit, not a measurement
model_static_polar_B_application.csv	17.2	101	10	lift, drag, moment and separation point of the model in a slow sweep at the case's Mach number: its quasi-steady limit, not a measurement
reconstruction_A_validation.csv	3.3	6	51	the reconstruction instant by instant: transpiration, circulations, the vortex marker and the pressure beneath it, extremes of the corrected pressure, of the local Mach number and of the temperatures, the size of the region beyond the critical pressure, closure of lift (corrected and incompressible), moment and drag
reconstruction_B_application.csv	3.3	6	51	the reconstruction instant by instant: transpiration, circulations, the vortex marker and the pressure beneath it, extremes of the corrected pressure, of the local Mach number and of the temperatures, the size of the region beyond the critical pressure, closure of lift (corrected and incompressible), moment and drag
response_surface.csv	2.7	42	11	peak lift, minimum moment, damping and onset incidence over mean incidence and reduced frequency
runtime_environment.csv	0.5	14	2	machine, library versions and CPU time of the march
static_station_spread.csv	0.4	4	7	Case B run with the static inputs of each Mach station in place of their blend
summary_all_cases.csv	0.5	2	13	one row of main results per case
time_history_A_validation.csv	436.3	3241	16	loads and model states at every step of the reported cycle
time_history_B_application.csv	584.8	4321	16	loads and model states at every step of the reported cycle